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ESC GUIDELINES

2026 ESC Guidelines for the management of heart failure

Supplementary data

Developed by the task force for the management of heart failure of the European Society of Cardiology (ESC)

With the special contribution of the Heart Failure Association (HFA) of the ESC

Endorsed by the European Association for Cardio-Thoracic Surgery (EACTS)

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Abbreviations and acronyms

6MWD	6-min walk distance
AF	Atrial fibrillation
ARNI	Angiotensin receptor–neprilysin inhibitor
ASCVD	Atherosclerotic cardiovascular disease
BMI	Body mass index
CABG	Coronary artery bypass grafting
CAD	Coronary artery disease
cAMP	cyclic adenosine monophosphate
CCS	Chronic coronary syndrome
CI	Confidence interval
COVID-19	Coronavirus disease 2019
CRT	cardiac resynchronization therapy
CV	Cardiovascular
FMT	Foundational medical therapy
HCM	Hypertrophic cardiomyopathy
HF	Heart failure
HFH	Heart failure hospitalization
HF-MDT	Heart failure multidisciplinary team
HF-MP	Heart failure management programme
HFpEF	Heart failure with preserved ejection fraction
HFrEF	Heart failure with reduced ejection fraction
HR	Hazard ratio
i.v.	Intravenous
ICD	Implantable cardioverter-defibrillator
ID	Iron deficiency
LBBAP	Left bundle branch area pacing
LBBB	Left bundle branch block
LVEF	Left ventricular ejection fraction
LVOTO	Left ventricular outflow tract obstruction
MCS	Mechanical circulatory support
MI	Myocardial infarction
MR	Mitral regurgitation
M-TEER	Mitral transcatheter edge-to-edge repair
mWHO	Modified World Health Organization
NSVT	Non-sustained ventricular tachycardia
NT-proBNP	N-terminal pro-B-type natriuretic peptide
NYHA	New York Heart Association
PCI	Percutaneous coronary intervention
PCP	Primary care professional
PHQ	Patient Health Questionnaire
PPCM	Peri-partum cardiomyopathy
QoL	Quality of life
RCT	Randomized clinical trial
RR	Rate ratio/risk ratio
SCD	Sudden cardiac death
Trans-CV	Trans-coronary vein
TSAT	Transferrin saturation
WCD	Wearable cardioverter-defibrillator

Table S1 Trial acronyms

ABYSS	Assessment of Beta-Blocker Interruption 1 Year after an Uncomplicated Myocardial Infarction on Safety and Symptomatic Cardiac Events Requiring Hospitalization
ADVENT-HF	Effect of Adaptive Servo-ventilation on Survival and Cardiovascular Hospital Admissions in Patients with Heart Failure and Sleep Apnoea
ADVOR	Acetazolamide in Decompensated Heart Failure with Volume Overload
AFFIRM	Atrial Fibrillation Follow-up Investigation of Rhythm Management
AFFIRM-AHF	A Randomized, Double-blind Placebo Controlled Trial Comparing the Effect of Intravenous Ferric Carboxymaltose on Hospitalizations and Mortality in Iron Deficient Subjects Admitted for Acute Heart Failure
AMBU-VAD	LVAD vs GDMT in Ambulatory Advanced Heart Failure Patients
AMICA	Atrial Fibrillation Management in Congestive Heart Failure With Ablation
APOLLO-B	Patisiran Treatment in Patients with Transthyretin Cardiac Amyloidosis
ATTR-ACT	Transthyretin Amyloidosis Cardiomyopathy Clinical Trial
ATTRibute-CM	Efficacy and Safety of AG10 in Subjects with Transthyretin Amyloid Cardiomyopathy
BETAMI-DANBLOCK	Beta-Blockers after Myocardial Infarction in Patients without Heart Failure
BLOCK-HF	Biventricular vs Right Ventricular Pacing in Heart Failure Patients with Atrioventricular Block
BUDAPEST-CRT Upgrade	Biventricular Upgrade on left ventricular reverse remodelling and clinical outcomes in patients with left ventricular Dysfunction and intermittent or permanent APical/SepTal right ventricular pacing Upgrade CRT
CABANA	Catheter ABlation vs ANti-arrhythmic drug therapy for Atrial fibrillation
CAMERA-MRI	Catheter Ablation vs Medical Rate Control in Atrial Fibrillation and Systolic Dysfunction
CAPRICORN	Carvedilol Post-Infarct Survival Control in LV Dysfunction
CARE-HF	CArdiac REsynchronization in Heart Failure

Continued

CASTLE-AF	Catheter Ablation vs Standard Conventional Treatment in Patients with Left Ventricular Dysfunction and Atrial Fibrillation
CASTLE-HTx	Catheter Ablation for Atrial Fibrillation in Patients With End-Stage Heart Failure and Eligibility for Heart Transplantation
CATHEDRAL-HF	Carvedilol as single maintenance therapy for Heart Failure with Improved Ejection Fraction
CHAMP-HF	Change the Management of Patients with Heart Failure (Registry)
CHARM-Alternative	Candesartan in Heart failure: Assessment of Reduction in Mortality and morbidity (CHARM)-Alternative
CHARM-Preserved	Candesartan in Heart failure: Assessment of Reduction in Mortality and morbidity-Preserved
CLOROTIC	Safety and Efficacy of the Combination of Loop with Thiazide-type Diuretics in Patients with Decompensated Heart Failure
COACH	Comparison of Outcomes and Access to Care for Heart Failure
COAPT	Cardiovascular Outcomes Assessment of the MitraClip Percutaneous Therapy for Heart Failure Patients with Functional Mitral Regurgitation
COMPANION	Comparison of Medical Therapy, Pacing, and Defibrillation in Heart Failure
CONFIRM-HF	Ferric CarboxymaltOse evaluatioN on perFormance in patients with IRon deficiency in coMbination with chronic Heart Failure
CONTAK-CD	Randomized controlled trial of the CONTAK-CD device
CREDENCE	Canagliflozin and Renal Endpoints in Diabetes with Established Nephropathy Clinical Evaluation
DanGer Shock	Danish–German Cardiogenic Shock
DANISH	Danish Study to Assess the Efficacy of ICDs in Patients with Non-ischaemic Systolic Heart Failure on Mortality
DAPA-CKD	Dapagliflozin And Prevention of Adverse outcomes in Chronic Kidney Disease
DAPA-HF	Dapagliflozin And Prevention of Adverse outcomes in Heart Failure
DapaTAVI	Dapagliflozin in Patients Undergoing Transcatheter Aortic-Valve Implantation
DELIVER	Dapagliflozin Evaluation to Improve the LIVEs of Patients with PReserved Ejection Fraction Heart Failure
DIG	Digitalis Investigation Group
DOSE	Diuretic Optimization Strategies Evaluation
ECLS-SHOCK	Extracorporeal Life Support in infarct-related Cardiogenic Shock
EFFECT-HF	Effect of Ferric Carboxymaltose on Exercise Capacity in Patients with Iron Deficiency and Chronic Heart Failure
EMPA-KIDNEY	Study of Heart and Kidney Protection with Empagliflozin
EMPEROR-Preserved	Empagliflozin Outcome Trial in Patients with Chronic Heart Failure with Preserved Ejection
EMPEROR-Reduced	EMPagliflozin outcomE tRial in Patients with chrOnic heaRt Failure with Reduced Ejection Fraction
EMPHASIS-HF	Eplerenone in Mild Patients Hospitalization And SurvIval Study in Heart Failure
EMPULSE	Empagliflozin in Patients Hospitalized with Acute Heart Failure Who Have Been Stabilized
ENACT-HF	Efficacy of a Standardized Diuretic Protocol in Acute Heart Failure
EPHESUS	Eplerenone Post-Acute Myocardial Infarction Heart Failure Efficacy and Survival Study
ExTraMATCH II	Exercise Training Meta-Analysis of Trials in Chronic Heart Failure
FAIR-HF	Ferinject Assessment in Patients with Iron Deficiency and Chronic Heart Failure
FAIR-HF2	Intravenous Iron in Patients with Systolic Heart Failure and Iron Deficiency to Improve Morbidity and Mortality
FIDELIO-DKD	Finerenone in Reducing Kidney Failure and Disease Progression in Diabetic Kidney Disease
FIGARO-DKD	Finerenone in Reducing Cardiovascular Mortality and Morbidity in Diabetic Kidney Disease
FINEARTS-HF	FINerenone trial to investigate Efficacy and sAfety superioR to placebo in paTientS with Heart Failure
FLOW	Research Study to See How Semaglutide Works Compared to Placebo in People with Type 2 Diabetes and Chronic Kidney Disease
GALACTIC-HF	Global Approach to Lowering Adverse Cardiac Outcomes through Improving Contractility in Heart Failure
HEAAL	Heart failure Endpoint evaluation of Angiotensin II Antagonist Losartan
HEART-FID	Ferric Carboxymaltose in Heart Failure with Iron Deficiency
HELIOS-B	Vutrisiran in Patients with Transthyretin Amyloidosis with Cardiomyopathy
HF-ACTION	Participants in Heart Failure: A Controlled Trial Investigating Outcomes of Exercise Training
HF-OPT	Heart Failure Optimization Study
HOBIPACE	Homburg Biventricular Pacing Evaluation
IRONMAN	Effectiveness of Intravenous iron treatment vs standard care in patients with heart failure and iron deficiency
ISCHEMIA trial	International Study of Comparative Health Effectiveness with Medical and Invasive Approaches
LESSER EARTH	Evaluation of Resynchronization Therapy for Heart Failure

Continued

MADIT-CRT	Multicenter Automatic Defibrillator Implantation Trial with Cardiac Resynchronization Therapy
MADIT-II	Multicenter Automatic Defibrillator Implantation Trial II
MIRACLE-ICD	Multicentre InSync Randomized Clinical Evaluation
MITRA-FR	Percutaneous Repair or Medical Treatment for Secondary Mitral Regurgitation
MOMENTUM 3	Multicenter Study of MagLev Technology in Patients Undergoing Mechanical Circulatory Support Therapy with HeartMate 3
MUSTIC	Multisite Stimulation in Cardiomyopathies
MUSTIC-AF	Multisite Stimulation in Cardiomyopathies—Atrial Fibrillation
MUSTIC-SR	Multisite Stimulation in Cardiomyopathies—Sinus Rhythm
PARADIGM-HF	Prospective Comparison of ARNI with ACE-I to Determine Impact on Global Mortality and Morbidity in Heart Failure
PARAGON-HF	Prospective Comparison of ARNI with ARB Global Outcomes in HF with preserved Ejection Fraction
PATH-CHF	Pacing Therapies for Congestive Heart Failure
PATHWAY-2	Spironolactone vs placebo, bisoprolol, and doxazosin to determine the optimal treatment for drug-resistant hypertension
PEACE	Prevention of Events with Angiotensin-Converting Enzyme Inhibition
PREDIMED	Prevention with Mediterranean Diet
PEP-CHF	Perindopril in Elderly People with Chronic Heart Failure
PROSPECT	Predictors of Response to CRT
PUSH-AHF	Pragmatic Urinary Sodium-based treatment algoritHm in Acute Heart Failure
RACE	Rate Control vs Electrical cardioversion
RACE II	Rate Control Efficacy in Permanent Atrial Fibrillation
RAFT	Resynchronization-Defibrillation for Ambulatory Heart Failure trial
RAFT-AF	Randomized Ablation-Based Rhythm-Control vs Rate-Control Trial in Patients with Heart Failure and Atrial Fibrillation
RALES	Randomized Aldactone Evaluation Study
REALIZE-K	Study to Assess Efficacy and Safety of SZC for the Management of High Potassium in Patients with Symptomatic HFrEF Receiving Spironolactone
REBIRTH	Randomized Evaluation of Bromocriptine In Myocardial Recovery THerapy for Peripartum Cardiomyopathy
REBOOT-CNIC	Beta-Blockers after Myocardial Infarction without Reduced Ejection Fraction
RED-HF	Reduction of Events with Darbepoetin alfa in Heart Failure
REDUCE LAP-HF II	A Study to Evaluate the Corvia Medical Inc. IASD System II to Reduce Elevated Left Atrial Pressure in Patients with Heart Failure
Rehab-VAD	Cardiac Rehabilitation in Patients with Continuous Flow Left Ventricular Assist Devices
RELIEVE-HF	Reducing Lung Congestion Symptoms in Advanced Heart Failure
REMATCH	Randomized Evaluation of Mechanical Assistance for the Treatment of Congestive Heart Failure
RESHAPE-HF	Transcatheter Valve Repair in Heart Failure with Moderate to Severe Mitral Regurgitation
RESHAPE-HF2	Transcatheter Valve Repair in HF with Moderate to Severe Mitral Regurgitation
ReThinQ	Cardiac Resynchronization Therapy in Patients with Heart Failure and Narrow QRS
REVERSE	REsynchronization reVErses Remodeling in Systolic left vEntricular dysfunction
REVIVED-BCIS2	Study of Efficacy and Safety of Percutaneous Coronary Intervention to Improve Survival in Heart Failure
SAVE	Survival and Ventricular Enlargement
SCD-HeFT	Sudden Cardiac Death in Heart Failure Trial
SELECT	Semaglutide Effects on Cardiovascular Outcomes in People with Overweight or Obesity
SENECOR	Randomized Controlled Trial Comparing a Multidisciplinary Intervention by a Geriatrician and a Cardiologist to Usual Care after a Heart Failure Hospitalization in Older Patients
SEQUOIA-HCM	Safety, Efficacy, and Quantitative Understanding of Obstruction Impact of Aficamten in HCM
SERVE-HF	Treatment of Sleep-Disordered Breathing with Predominant Central Sleep Apnea by Adaptive Servo Ventilation in Patients with Heart Failure
SHARC	Shock Academic Research Consortium
SHIFT	Systolic Heart failure treatment with the I_f inhibitor ivabradine Trial
SOLOIST-WHF	Effect of Sotagliflozin on Cardiovascular Events in Patients with Type 2 Diabetes Post Worsening Heart Failure
SOLVD-Prevention	Studies of Left Ventricular Dysfunction—Prevention

Continued

SOLVD-Treatment	Studies of Left Ventricular Dysfunction—Treatment
STEP-HFpEF	Effect of Semaglutide 2.4 mg Once Weekly on Function and Symptoms in Subjects with Obesity-related Heart Failure with Preserved Ejection Fraction
STEP-HFpEF DM	Semaglutide Treatment Effect in People with Obesity and Heart Failure with Preserved Ejection Fraction and Diabetes Mellitus
STICH	The Coronary-Artery Bypass Surgery in Patients with Left Ventricular Dysfunction
STOP-CA	Statins to Prevent the Cardiotoxicity of Anthracyclines
STOP-CRT	Systematic Withdrawal of Neurohumoral Blocker Therapy in Optimally Responding CRT Patients
STRONG-HF	Safety, Tolerability and Efficacy of Rapid Optimization, Helped by NT-proBNP Testing, of Heart Failure Therapies
SUMMIT	A Study of Tirzepatide in Participants with Heart Failure with Preserved Ejection Fraction and Obesity
SweVAD	SWEdish evaluation of left Ventricular Assist Device
TEAM-HF	Trial for Evaluating Accelerated Guideline-Directed Medical Therapy for Heart Failure
TOPCAT	Treatment of Preserved Cardiac Function Heart Failure with an Aldosterone Antagonist
TRACE	Trandolapril Cardiac Evaluation
TRED-HF	Withdrawal of pharmacological treatment for heart failure in patients with recovered dilated cardiomyopathy
Tri.fr	Transcatheter Edge-to-Edge Repair for Severe Isolated Tricuspid Regurgitation
TRILUMINATE Pivotal	Trial to Evaluate Cardiovascular Outcomes in Patients Treated with the Tricuspid Valve Repair System Pivotal
TRISCEND II	EVOQUE Transcatheter Tricuspid Valve Replacement: Pivotal Clinical Investigation of Safety and Clinical Efficacy Using a Novel Device
Val-HeFT	Valsartan Heart Failure Trial
VEST	Vest Prevention of Early Sudden Death
VICTORIA	Vericiguat Global Study in Patients with Heart Failure with Reduced Ejection Fraction

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1. Prevention of heart failure

Four different risk scores have been developed by the European Society of Cardiology (ESC) to aid in heart failure (HF) risk stratification (Supplementary data online, *Figure S1*). Two of these (SCORE2-HF and SMART2-HF) are intended to estimate the risk of incident HF in people without HF.[1,2] The other two scores (LIFE-Preserved and LIFE-HF) estimate the risk of HF hospitalization (HFH) and cardiovascular (CV) death in patients with HF with preserved ejection fraction (HFpEF) and HF with reduced ejection fraction (HFrEF), respectively.[3,4]

1.1. Risk scores for individuals without heart failure

1.1.1. SCORE2-HF

SCORE2-HF is intended for use in a primary preventive setting (stage A or B HF) among people without a history of atherosclerotic CV disease (ASCVD).[1] It estimates the 10-year risk of fatal or non-fatal HF (by sex) according to the following risk factors: age, systolic blood pressure, body mass index (BMI), estimated glomerular filtration rate, current use of antihypertensive medication, current smoking, history of diabetes, and (for those with diabetes) age at onset of diabetes and current glycated haemoglobin. Because the absolute 10-year risk tends to be low even in the presence of multiple risk factors among young adults, 30-year risks can also be estimated to better reflect the long-term risk of HF to guide early preventive strategies.[1] We do not recommend the initiation of a certain therapy at a specific risk threshold, but rather this score is intended to be used as a tool to facilitate discussion of prevention between the patient and the physician.

1.1.2. SMART2-HF

SMART2-HF is intended for use in a primary preventive setting (stage A or B HF) among people with a history of ASCVD.[2] It estimates the 10-year risk of fatal or non-fatal HF (by sex) according to the following risk factors: age, coronary artery disease (CAD), cerebrovascular disease, peripheral artery disease, abdominal aortic aneurysm, atrial fibrillation (AF), diabetes, current smoking, systolic blood pressure, BMI, C-reactive protein concentration, estimated glomerular filtration rate, non-high-density lipoprotein cholesterol concentration, and years since first diagnosis of ASCVD.[2] We do not recommend the initiation of a certain therapy at a specific risk threshold, but rather this score is intended to be used as a tool to facilitate discussion of prevention between the patient and the physician.

1.2. Risk prediction in patients with established HF

LIFE-Preserved predicts the risk of HFH or CV death in patients with pre-existing HFpEF [i.e. stage C or D HF, and a left ventricular ejection fraction (LVEF) ≥50%].[3] Two- to 10-year risks as well as lifetime risks can be calculated using the following factors: sex, age, current smoking, New York Heart Association (NYHA) class, ischaemic heart disease, chronic obstructive pulmonary disease, AF, prior HFH (at any time), HFH within the last 6 months, diabetes, N-terminal pro-B-type natriuretic peptide (NT-proBNP) level, BMI, haemoglobin concentration, estimated glomerular filtration rate, and heart rate.[3] As for the other scores, we do not recommend the initiation of a certain therapy at a specific risk threshold, but rather this score is intended to be used as a tool to facilitate discussion of various treatment options between the patient and the physician.

1.3. LIFE-HF

LIFE-HF predicts the risk of HFH or CV death in patients with pre-existing HFrEF and LVEF ≤40% (i.e. stage C or D HF).[4] Two- to 10-year risks as well as lifetime risks can be calculated using the following factors: sex, age, NYHA class, prior HFH (at any time), diabetes, extracardiac vascular disease, systolic blood pressure, LVEF, NT-proBNP level, and estimated glomerular filtration rate.[4] As for the other scores, we do not recommend the initiation of a certain therapy at a specific risk threshold, but rather this score is intended to be used as a tool to facilitate discussion of various treatment options between the patient and the physician.

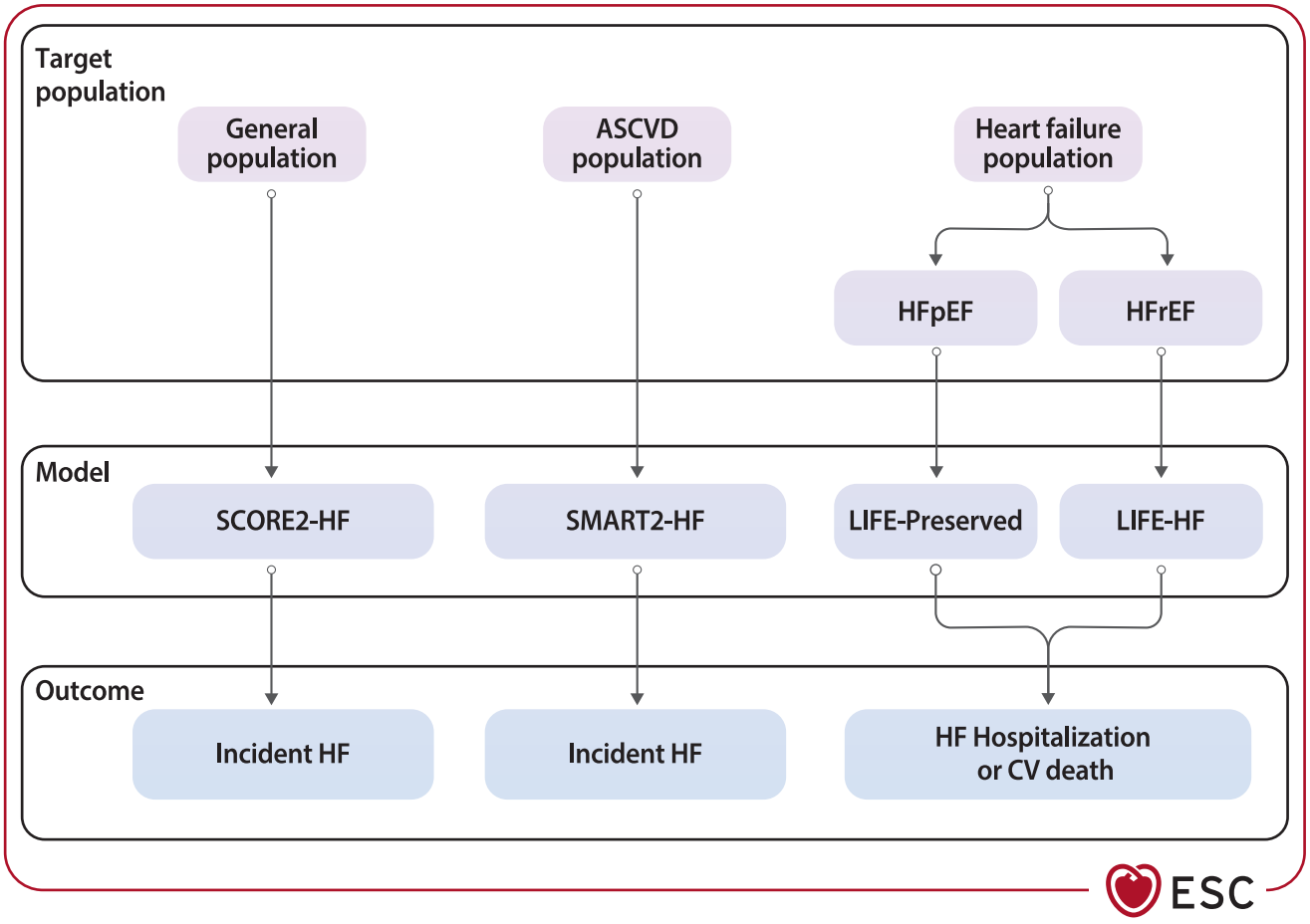


Figure S1 Overview of ESC heart failure risk scores. ASCVD, atherosclerotic cardiovascular disease; CV, cardiovascular; HF, heart failure; HFpEF, heart failure with preserved ejection fraction; HFrEF, heart failure with reduced ejection fraction.

2. Diagnosis of chronic heart failure

Table S2 Further testing depending on phenotype identified initially

Phenotype/aetiology	Aetiology subtype	Subsequent testing
Coronary artery disease	Ischaemic HF	CTCA Invasive coronary angiography
Valvular heart disease	Primary valve disease Secondary valve disease	Transoesophageal echocardiography CMR
Atrial fibrillation/ flutter/other SVTs	Arrhythmia-associated HF	Ambulatory ECG Continuous ECG Repeat TTE in sinus rhythm
Ventricular arrhythmia	Arrhythmia-associated HF	Ambulatory ECG, EPS if indicated
Pacing- or LBBB	HF associated with abnormal cardiac conduction	ECG, device interrogation
Cardiomyopathies	DCM/HCM/RCM/ARVC/NDLVC	CMR, genetic testing
Infiltrative cardiomyopathy	AL-CM or ATTR-CM	Paraprotein testing, DPD/PYP/HMPD scintigraphy, ±genetic testing, ±extracardiac biopsy, ±EMB
Sarcoid	Sarcoid	Chest CT, 18F-FDG-PET, CMR, ACE level, IL-2, ambulatory ECG, ±extracardiac biopsy, ±EMB
Takotsubo syndrome	Takotsubo syndrome	CMR, ICA or CTCA in selected cases, repeat TTE
Cancer therapy-induced	Anthracycline chemotherapy	NP, TTE with GLS
	HER2-targeted therapy	NP, TTE with GLS
	Immune checkpoint inhibitors	cTn, NP, TTE, CMR
	CAR-T therapies	Continuous ECG, TTE, cTn, NP, IL-6 rise is supportive
Drug-induced	Toxins: alcohol, cocaine, steroids, stimulants	Toxicology if indicated
Post infective	Viral myocarditis HIV	Viral screening, PCR, CMR (±EMB)
	Chagas disease	*Trypanosoma cruzi* antibody testing (ELISA, RIPA)
	HIV	Serology
Storage diseases	Iron overload CM	T2* CMR
	Fabry disease	CMR, α-galactosidase A, genetics
	Glycogen storage disease	CMR, genetic testing
Endomyocardial disease	Endomyocardial fibrosis	EMB/skin biopsy
	Carcinoid	24 h urine 5-HIAA
Metabolic disease	Endocrine dysfunction	Endocrinology tests
	Nutritional deficiency	Specific nutrient testing
Autoimmune disease	Large vessel vasculitis	ANCA, ESR, CT-PET, rheumatology review
	Granulomatosis with polyangiitis (Wegener's)	CRP, ANCA, chest CT, rheumatology review
	SLE	ANA, ESR/CRP
	Eosinophilic granulomatosis with polyangiitis	Full blood count including eosinophilic count, ESR/CRP, ANCA, CMR, chest CT, pulmonary function tests, extracardiac biopsy ±EMB, rheumatology review
Neuromuscular disease	Becker/Duchenne/Emery–Dreifuss muscular dystrophy Friedrich's ataxia	CK, EMG, genetics Nerve conduction studies, EMG, genetics
Pericardial disease	Constrictive pericarditis	Chest CT, CMR, L+RHC

18F-FDG-PET, 18F-fluorodeoxyglucose-positron emission tomography; 5-HIAA, 5-hydroxyindoleacetic acid; ACE, angiotensin-converting enzyme; AL-CM, immunoglobulin light chain amyloid cardiomyopathy; ANA, anti-nuclear antibody; ANCA, anti-neutrophil cytoplasmic antibody; ARVC, arrhythmogenic right ventricular cardiomyopathy; ATTR-CM, transthyretin amyloid cardiomyopathy; CAR-T, chimeric antigen receptor T-cell, CK, creatine kinase; CM, cardiomyopathy; CMR, cardiac magnetic resonance; CRP, C-reactive protein; CT, computed tomography; CTCA, computed tomography coronary angiography; cTn, cardiac troponin; DCM, dilated cardiomyopathy; DPD, 3,3-diphosphono-1,2-propanodicarboxylic acid; ECG, electrocardiogram; ELISA, enzyme-linked immunosorbent assay; EMB, endomyocardial biopsy; EMG, electromyography; EPS, electrophysiology study; ESR, erythrocyte sedimentation rate; FDG-PET, fluorodeoxyglucose–positron emission tomography; GLS, global longitudinal strain; HCM, hypertrophic cardiomyopathy; HER2, human epidermal growth factor receptor 2; HF, heart failure; HIV, human immunodeficiency virus; HMPD, hydroxymethylene diphosphonate; ICA, invasive coronary angiography; IL, interleukin; L+RHC, left and right heart catheterization; LBBB, left bundle branch block; NDLVC, non-dilated left ventricular cardiomyopathy; NP, natriuretic peptide; NSVT, non-sustained ventricular tachycardia; PCR, polymerase chain reaction; PET, positron emission tomography; PYP, pyrophosphate; RCM, restrictive cardiomyopathy; RIPA, radioimmunoprecipitation; SLE, systemic lupus erythematosus; SVT, supraventricular tachycardia; T2*, T2 star relaxometry; TTE, transthoracic echocardiography.

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3. Management of chronic heart failure

3.1. Pharmacological therapy

Table S3 Main characteristics of different mineralocorticoid receptor antagonists

	Spironolactone	Eplerenone	Finerenone
Generation	First (steroidal)	Second (steroidal)	Non-steroidal
Dosing	12.5–50 mg o.d.	25–50 mg o.d.	10–40 mg o.d.
Caution[a]	Potassium increase >5.5 mmol/L If eGFR decreases <30 mL/min/1.73 m^2	Potassium increase >5.5 mmol/L If eGFR decreases <30 mL/min/1.73 m^2	Potassium increase >5.5 mmol/L If eGFR decreases <15 mL/min/1.73 m^2 or eGFR decreases >40%
Receptor selectivity	Blocks mineralocorticoid + androgen/progesterone receptors	More selective for mineralocorticoid receptors	Highly selective for mineralocorticoid receptors
Hyperkalaemia risk	Moderate to high	Moderate to high	Low to moderate
Clinical HF trials[b,c]	RALES (1999)—HFrEF[5] TOPCAT (2014)—HFpEF[6]	EMPHASIS-HF (2011)—HFrEF[7]	FINEARTS-HF (2024)—HFpEF[8]
Clinical trials for non-HF endpoints[b]	PATHWAY-2 (2015)—hypertension[9]	EPHESUS (2003)—LV dysfunction post-MI[10]	FIDELIO-DKD (2020),[11] FIGARO-DKD (2021)[12]—kidney and CV protection in CKD/T2DM
Metabolism	Hepatic (CYP3A4)	Hepatic (CYP3A4)	Hepatic (CYP3A4)
Elimination	Renal and hepatic	Renal and hepatic	Mostly hepatic
Main side effects	Gynaecomastia, hyperkalaemia	Gynaecomastia, hyperkalaemia	Hyperkalaemia

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CKD, chronic kidney disease; CV, cardiovascular; CYP3A4, cytochrome P450 isoenzyme 3A4; eGFR, estimated glomerular filtration rate; HF, heart failure; HFpEF, heart failure with preserved ejection fraction; HFrEF, heart failure with reduced ejection fraction; LV, left ventricular; MI, myocardial infarction; o.d., once daily; T2DM, type 2 diabetes mellitus.
[a]See Supplementary data online, *Table S8*.
[b]See Supplementary data online, *Table S1* for acronyms of trials.
[c]For further information, see Supplementary data online, *Evidence Tables 32* and *35*.

Table S4 Practical guidance on the use of angiotensin-converting enzyme inhibitors (or an angiotensin II receptor blockers) in patients with heart failure

WHY?
To improve symptoms and exercise capacity, reduce the risk of HF hospitalization and increase survival.
IN WHOM AND WHEN?
(1) All patients with HF, irrespective of LVEF. (2) ACE-Is are also of benefit in patients with asymptomatic LV systolic dysfunction (NYHA class I).
Contraindications: (1) History of angio-oedema.[a] (2) Known bilateral renal artery stenosis. (3) Pregnancy/risk of pregnancy and breastfeeding period. (4) Known allergic reaction/other adverse reaction (drug-specific).
Cautions/seek specialist advice: (1) Significant hyperkalaemia (K^+ >5.2 mmol/L). (2) Significant renal dysfunction (eGFR <30 mL/min/1.73 m^2). May be considered in lower eGFR under close monitoring of eGFR after initiation. (3) Symptomatic or severe asymptomatic hypotension (systolic blood pressure <90 mmHg). (4) Drug interactions to look out for: • K^+ supplements K^+-sparing diuretics, e.g. amiloride and triamterene (beware combination preparations with furosemide). • MRAs.

Continued

- Renin inhibitors.[b]
- NSAIDs.[c]
- Trimethoprim/trimethoprim-sulfamethoxazole.
- 'Low-salt' substitutes with a high K^+ content.

WHICH ACE-I AND WHAT DOSE?[d]

Captopril: starting dose 6.25 mg t.i.d., target dose 50 mg t.i.d.
Enalapril: starting dose 2.5 mg b.i.d., target dose 10–20 mg b.i.d.
Lisinopril: starting dose 2.5–5.0 mg o.d., target dose 20–35 mg o.d.
Ramipril: starting dose 1.25–2.5 mg b.i.d., target dose 5 mg b.i.d.
Trandolapril: starting dose 0.5 mg o.d., target dose 4 mg o.d.
If ACE-I is not tolerated due to ACE-I-specific adverse events and ARB can be used:
Candesartan: starting dose 4 mg o.d., target dose 32 mg o.d.
Valsartan: starting dose 40 mg b.i.d., target dose 160 mg b.i.d.
Losartan: starting dose (12,5) 25–50 mg o.d., target dose 150 mg o.d.

WHERE?

- In the community in stable patients (NYHA class IV/patients with advanced HF and those with a current/recent exacerbation should be referred for specialist advice).
- In patients hospitalized with DHF—after stabilizing (but ideally before discharge).
- Other exceptions—see '**Cautions/seek specialist advice**'.

HOW TO USE?

- Check renal function and electrolytes.
- Start with a low dose (see above[d]).
- Rapid dose uptitration may be carried out in patients in hospital or who are otherwise closely monitored, tolerability permitting. In those patients where close monitoring is not possible, dose should be doubled at not less than 2-week intervals.
- Aim for the target dose (see above) or, failing that, the highest tolerated dose [remember: some ACE-I (or ARB) is better than no ACE-I].
- Recheck blood chemistry (urea/BUN, creatinine, K^+) 1–2 weeks after initiation and 1–2 weeks after final dose titration.
- Monitor blood chemistry 4–6 monthly thereafter.
- When to stop uptitration, reduce dose, stop treatment—see PROBLEM SOLVING.
- It is very rarely necessary to stop an ACE-I (or ARB), and clinical deterioration is likely if treatment is withdrawn. Ideally, specialist advice should be sought before treatment discontinuation.
- A specialist HF nurse may assist with education of the patient, follow-up (in person or by telephone), biochemical monitoring, and dose uptitration.

If ACE-I (or ARB) has been temporarily discontinued for less than 14 days, it should, in most cases, be safe to initiate directly at the previously tolerated doses in stable patients.

PROBLEM SOLVING

Asymptomatic low blood pressure:

- Does not usually require any change in therapy.

Symptomatic hypotension:

- Dizziness/light headedness is common and often improves over time—patients should be reassured.
- Reconsider need for nitrates, calcium channel blockers,[e] and other vasodilators and reduce dose/stop, if possible.
- If no signs or symptoms of congestion, consider reducing diuretic dose.
- If these measures do not solve problem, seek specialist advice.

Cough:

- Cough is common in patients with HF, many of whom have smoking-related lung disease.
- Cough is also a symptom of pulmonary oedema, which should be excluded when a new worsening cough develops.
- ACE-I-induced cough does not always require treatment discontinuation.
- When a troublesome cough does develop (e.g. one stopping the patient from sleeping) and can be proved to be due to ACE inhibition (i.e. recurs after ACE-I withdrawal and rechallenge), substitution with an ARB is recommended.

Worsening renal function and hyperkalaemia:

- Some rise in urea (BUN), creatinine, and potassium is to be expected after an ACE-I; if an increase is small, no action is necessary.
- A decrease in eGFR of <30% is acceptable.
- An increase in potassium to ≤5.5 mmol/L is acceptable.
- If urea, creatinine, or potassium does rise excessively, consider stopping concomitant nephrotoxic drugs (e.g. NSAIDs[c]) and other potassium supplements or retaining agents (triamterene, amiloride) and, if no signs of congestion, reducing the dose of diuretic.

Continued

- If greater rises in creatinine or potassium than those outlined above persist despite adjustment of concomitant medications, the dose of the ACE-I (or ARB) should be halved and blood chemistry rechecked within 1–2 weeks; if there is still an unsatisfactory response, specialist advice should be sought.
- If potassium rises to >5.5 mmol/L or creatinine increases by >100% or to eGFR <20 mL/min/1.73 m^2, the ACE-I (or ARB) should be stopped and specialist advice sought.
- Blood chemistry should be monitored frequently and serially until potassium and creatinine have plateaued.

ADVICE TO THE PATIENT

- Explain expected benefits:
  - Improved symptoms and exercise capacity.
  - Prevention of worsening of HF leading to hospital admission.
  - Increased survival.
- Symptoms improve within a few weeks to a few months after starting treatment.
- Advise patients to report principal adverse effects (i.e. dizziness/symptomatic hypotension, cough)—see PROBLEM SOLVING.
- Advise patients to avoid NSAIDs[c] not prescribed by a physician (i.e. purchased over-the-counter) and salt substitutes high in K^+—see PROBLEM SOLVING.

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ACE-I, angiotensin-converting enzyme inhibitor; ARB, angiotensin II receptor blocker; b.i.d., twice daily; BUN, blood urea nitrogen; DHF, Decompensated Heart Failiure; eGFR, estimated glomerular filtration rate; HF, heart failure; HFrEF, heart failure with reduced ejection fraction; LV, left ventricular; LVEF, left ventricular ejection fraction; MRA, mineralocorticoid receptor antagonist; NSAID, non-steroidal anti-inflammatory drug; NYHA, New York Heart Association; o.d., once daily; t.i.d., three times a day.
[a]The safety of an ARB in patients developing angio-oedema with an ACE-I is uncertain.
[b]Renin inhibitors are not recommended in HF.
[c]Avoid NSAIDs unless essential.
[d]See main guidelines, *Table 11*.
[e]Calcium channel blockers should be discontinued unless absolutely necessary, and diltiazem and verapamil are potentially harmful in HFrEF because of their negative inotropic action.
The advice in this table represents expert opinion based upon relevant clinical trials (drugs, titration schedules, target doses, patient monitoring, treatment benefits, and reported adverse effects) and clinical experience.
For further details on class of recommendation and level of evidence, see main guidelines *Sections 4* and *6* as well as Supplementary data online, *Evidence Tables 29–41*.

Table S5 Practical guidance for the use of angiotensin receptor–neprilysin inhibitors in patients with heart failure

WHY?

To improve symptoms, reduce the risk of HF hospitalization and increase survival.

IN WHOM AND WHEN?

(1) All patients with HF and persisting symptoms (NYHA classes II–IV), irrespective of LVEF.

Contraindications:

(1) History of angio-oedema.[a]
(2) Known bilateral renal artery stenosis.
(3) Pregnancy/risk of pregnancy and breastfeeding period.
(4) Known allergic reaction/other adverse reaction (drug-specific).
(5) Symptoms of hypotension or a systolic blood pressure <90 mmHg (PARADIGM-HF enrolled patients with a systolic blood pressure >95 mmHg at randomization).

Cautions/seek specialist advice:

(1) A washout period of at least 36 h after ACE-I therapy is required in order to minimize the risk of angio-oedema.
(2) Significant hyperkalaemia (K^+ >5.2 mmol/L).
(3) eGFR <30 mL/min/1.73 m^2.
(4) Drug interactions to look out for:
  - K^+ supplements/K^+-sparing diuretics, e.g. amiloride and triamterene (beware combination preparations with furosemide).
  - MRAs.
  - Renin inhibitors.[b]
  - NSAIDs.[c]
  - Trimethoprim/trimethoprim-sulfamethoxazole.
  - 'Low-salt' substitutes with a high K^+ content.

WHAT DOSE?[d]

- ACE-I/ARB naïve patients: Starting dose 24/26 or 49/51 mg b.i.d., depending on clinical assessment. Target dose 97/103 mg b.i.d.

Continued

- On target dose of ACE-I/ARB: Starting dose 49/51 mg b.i.d., target dose 97/103 mg b.i.d.
- Not on target dose of ACE-I/ARB: Starting dose 24/26 mg b.i.d., target dose 97/103 mg b.i.d.

WHERE?

- In the community in stable patients (NYHA class IV/patients with advanced HF and those with a current/recent exacerbation should be referred for specialist advice).
- In patients hospitalized with DHF—after stabilizing, relieving congestion, and if possible, restoring 'euvolaemia' (but ideally before discharge).
- Other exceptions—see 'Cautions/seek specialist advice'.

HOW TO USE?

- Check renal function and electrolytes prior to initiation.
- Start with a low dose (see above[d]).
- In some patients, one may consider a reduced starting dose (24/26 mg b.i.d.), namely in those with systolic blood pressure 100–110 mmHg, ACE-I/ARB naïve patients, and in those with CKD.
- Rapid dose uptitration may be carried out in patients in hospital or who are otherwise closely monitored, tolerability permitting. In those patients where close monitoring is not possible, dose should be doubled at not less than 2-week intervals.
- Aim for the target dose (see above) or, failing that, the highest tolerated dose.
- Recheck blood chemistry (urea/BUN, creatinine, K^+) 1–2 weeks after initiation and 1–2 weeks after final dose titration.
- Monitor blood chemistry 4–6 monthly thereafter.
- When to stop uptitration, reduce dose, stop treatment—see PROBLEM SOLVING.
- It is very rarely necessary to stop an ARNI, and clinical deterioration is likely if treatment is withdrawn. Ideally, specialist advice should be sought before treatment discontinuation.
- A specialist HF nurse may assist with education of the patient, follow-up (in person or by telephone), biochemical monitoring, and dose uptitration.
- If ARNI has been temporarily discontinued for less than 14 days, it should, in most cases, be safe to initiate directly at the previously tolerated doses in stable patients.

PROBLEM SOLVING

Asymptomatic low blood pressure:

- Does not usually require any change in therapy.

Symptomatic hypotension:

- Dizziness/light headedness is common and often improves with time—patients should be reassured.
- Reconsider need for any other vasodilators and reduce dose/stop, if possible.
- If no signs or symptoms of congestion, consider reducing diuretic dose.
- If these measures do not solve problem, seek specialist advice.

Cough:

- Cough is common in patients with HF, many of whom have smoking-related lung disease.
- Cough is also a symptom of pulmonary oedema, which should be excluded when a new worsening cough develops.
- When a troublesome cough does develop (e.g. one stopping the patient from sleeping) and can be proved to be due to ARNI and ACE inhibition (i.e. recurs after the drugs withdrawal and rechallenge), substitution of an ARB is recommended.

Worsening renal function and hyperkalaemia:

- Some rise in urea (BUN), creatinine, and potassium is to be expected after an ARNI; if an increase is small, no action is necessary.
- A decrease in eGFR of <30% is acceptable.
- An increase in potassium up to ≤5.5 mmol/L is acceptable.
- If urea, creatinine, or potassium does rise excessively, consider stopping concomitant nephrotoxic drugs (e.g. NSAIDs[c]) and other potassium supplements or retaining agents (triamterene, amiloride), and, if no signs of congestion, reducing the dose of diuretic. This is particularly true in those patients on an SGLT2-I.
- If greater rises in creatinine or potassium than those outlined above persist despite adjustment of concomitant medications, the dose of the ARNI should be halved and blood chemistry rechecked within 1–2 weeks; if there is still an unsatisfactory response, specialist advice should be sought.
- Blood chemistry should be monitored frequently and serially until potassium and creatinine have plateaued.

ADVICE TO THE PATIENT

Explain expected benefits:

- Improved symptoms.
- Prevention of worsening of HF leading to hospital admission.
- Increased survival (reduction in both CV and all-cause mortality).

Continued

- Symptoms improve within a few weeks to a few months after starting treatment.
- Advise patients to report principal adverse effects (i.e. dizziness/symptomatic hypotension, cough)—see PROBLEM SOLVING.
- Advise patients to avoid NSAIDs[c] not prescribed by a physician (i.e. purchased over-the-counter) and salt substitutes high in K^+—see PROBLEM SOLVING.

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ACE-I, angiotensin-converting enzyme inhibitor; ARB, angiotensin II receptor blocker; ARNI, angiotensin receptor–neprilysin inhibitor; b.i.d., twice daily; BUN, blood urea nitrogen; eGFR, estimated glomerular filtration rate; HF, heart failure; LVEF, left ventricular ejection fraction; MRA, mineralocorticoid receptor antagonist; NSAID, non-steroidal anti-inflammatory drug; NYHA, New York Heart Association; SGLT2-I, sodium–glucose co-transporter 2 inhibitor.
[a]The safety of an ARB/ARNI in patients developing angioedema with an ACE-I is uncertain.
[b]Renin inhibitors are not recommended in HF.
[c]Avoid NSAIDs unless essential.
[d]See main guidelines Table 11.
The advice in this table represents expert opinion based upon relevant clinical trials (drugs, titration schedules, target doses, patient monitoring, treatment benefits, and reported adverse effects) and clinical experience.
For further details on class of recommendation and level of evidence, see main guidelines *Section 6* as well as Supplementary data online, *Evidence Tables 29–41*.

Table S6 Practical guidance for the use of beta-blockers in patients with heart failure

WHY?
To improve symptoms, reduce the risk of HF hospitalization and increase survival.
IN WHOM AND WHEN?
(1) All patients with stable HFrEF (and asymptomatic LV dysfunction). (2) Patients with advanced HF also benefit, but treatment should be started under the care of a specialist.
Contraindications: (1) Second- or third-degree AV block (in the absence of a permanent pacemaker). (2) Critical limb ischaemia. (3) Asthma (relative contraindication): if cardio-selective beta-blockers are indicated, asthma is not an absolute contraindication, but these medications should only be used under close medical supervision by a specialist, with consideration of the risks for and against their use; COPD is not a contraindication. (4) Known allergic reaction/other adverse reaction (drug-specific).
Cautions/seek specialist advice: (1) Advanced HF with NYHA class IV. (2) Current or recent (<4 weeks) exacerbation of HF (e.g. hospital admission with worsening HF), heart block, or heart rate <50 b.p.m. (3) If persisting signs of congestion, hypotension (systolic <90 mmHg), raised jugular venous pressure, ascites, marked peripheral oedema—try to relieve congestion and achieve 'euvolaemia' before starting a beta-blocker. (4) Drug interactions to look out for (because of risk of bradycardia/AV block): • Verapamil, diltiazem (are not recommended and should be discontinued).[a] • Digoxin. • Amiodarone. • Ivabradine.
WHICH BETA-BLOCKER AND WHAT DOSE?[b]
Bisoprolol: starting dose 1.25 mg o.d., target dose 10 mg o.d. Carvedilol: starting dose 3.125 mg b.i.d., target dose 25 mg b.i.d. (target dose 50 mg b.i.d. if >85 kg). Metoprolol succinate (CR/XL): starting dose 12.5–25 mg o.d., target dose 200 mg o.d. Nebivolol: starting dose 1.25 mg o.d., target dose 10 mg o.d.
WHERE?
• In the community in stable patients. • In patients hospitalized with DHF/WHF—after stabilizing, relieving congestion, and, if possible, restoring 'euvolaemia' (but ideally before discharge). • In patients with NYHA class IV/patients with advanced HF and those with a current/recent exacerbation, specialist advice should be sought. • Other exceptions—see 'Cautions/seek specialist advice'.
HOW TO USE?
• In general patients should be euvolaemic and in a stable condition prior to initiating treatment.

Continued

- Start with a low dose.[b]
- Rapid dose uptitration may be carried out in patients in hospital or who are otherwise closely monitored, tolerability permitting. In those patients where close monitoring is not possible, dose should be doubled at not less than 2-week intervals (slower uptitration may be needed in some patients).
- Aim for the target dose[b] or, failing that, the highest tolerated dose (remember: some beta-blocker is better than no beta-blocker).
- If a CIED is implanted (mainly CRT) and the main reason for not reaching target dose of beta-blocker is bradycardia, beta-blocker dose can often be increased after CIED implantation (beware of inappropriate RV pacing—see PROBLEM SOLVING).
- In cases of unacceptable non-cardiac side effect on low doses, consider trying another type of beta-blocker, as side effects differ between beta-blockers.
- Monitor heart rate, blood pressure, and clinical status (symptoms, signs—especially signs of congestion, body weight).
- A specialist HF nurse may assist with education of the patient, follow-up (in person or by telephone), and dose uptitration.
- When to stop uptitration, reduce dose, stop treatment—see PROBLEM SOLVING.
- If beta-blocker therapy has been temporarily discontinued for less than 14 days, it should, in most cases, be safe to initiate directly at the previously tolerated doses in stable patients. Euvolaemia is crucial prior to initiating beta-blockers.

PROBLEM SOLVING

General:

- Beta-blockers should not be stopped suddenly unless absolutely necessary (there is a risk of a 'rebound' increase in myocardial ischaemia or infarction and arrhythmias). Ideally, specialist advice should be sought before treatment discontinuation.

Worsening symptoms or signs (e.g. increasing dyspnoea, fatigue, oedema, weight gain):

- If increasing congestion, increase dose of diuretic or halve a dose of beta-blocker (if increasing diuretic dose does not work).
- If marked fatigue (or bradycardia—see below), halve a dose of beta-blocker (rarely necessary); review patient in 1–2 weeks; if not improved, seek specialist advice.
- If serious deterioration, halve a dose of beta-blocker or stop this treatment (rarely necessary); seek specialist advice.

Low heart rate:

- If <50 b.p.m. and worsening symptoms, lower dose of beta-blocker, or, if severe deterioration, stop beta-blocker (rarely necessary).
- Review need for other heart rate-slowing drugs (e.g. digoxin, ivabradine, amiodarone, diltiazem, or verapamil[a]).
- Arrange electrocardiogram to exclude heart block.
- Seek specialist advice.
- In patients with ICD-only or pacemakers, one should be vigilant not to increase RV pacing on the grounds of reaching target doses of beta-blockers. In cases of high RV pacing, reassess beta-blocker doses and seek expert advice.

Asymptomatic low blood pressure:

- Does not usually require any change in therapy.

Symptomatic hypotension:

- If dizziness, lightheadedness, or confusion and a low blood pressure, reconsider need for nitrates, calcium channel blockers,[a] and other vasodilators and reduce/stop, if possible.
- If no signs or symptoms of congestion, consider reducing diuretic dose.
- If these measures do not solve problem, seek specialist advice.

ADVICE TO THE PATIENT

- Explain expected benefits (see WHY?) and mention possibility of temporary adverse effects.
- Treatment is given to improve symptoms, to prevent worsening of HF leading to hospital admission, and to increase survival.
- Symptomatic improvement may develop slowly after starting treatment, sometimes taking 3–6 months or longer.
- Temporary symptomatic deterioration may occur during initiation or uptitration phase; in the long-term beta-blockers improve well-being.
- Advise patient to report deterioration (see PROBLEM SOLVING) and that deterioration (tiredness, fatigue, breathlessness) can usually be easily managed by an adjustment of other medication; patients should be advised not to stop beta-blocker therapy without consulting the physician.
- During initiation or uptitration phase to detect and to treat potential deterioration early, patients should be encouraged to weigh themselves daily (after waking, before dressing, after voiding, before eating), and to increase their diuretic dose should their weight increase, persistently (>2 days) by >1.5–2.0 kg/day.

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AV, atrioventricular; b.i.d., twice daily; CIED, cardiac implantable electronic device; COPD, chronic obstructive pulmonary disease; CR, controlled release; CRT, cardiac resynchronization therapy; HF, heart failure; HFrEF, heart failure with reduced ejection fraction; ICD, implantable cardioverter-defibrillator; LV, left ventricular; NYHA, New York Heart Association; o.d., once daily; RV, right ventricular; XL, extended release.

[a]Calcium channel blockers should be discontinued unless absolutely necessary, and diltiazem and verapamil are potentially harmful in HFrEF because of their negative inotropic effect.

[b]See main guidelines, Table 11.

The advice in this table represents expert opinion based upon relevant clinical trials (drugs, titration schedules, target doses, patient monitoring, treatment benefits, and reported adverse effects) and clinical experience.

For further details on class of recommendation and level of evidence, see the main guidelines *Section 6* as well as Supplementary data online, *Evidence Tables 29–41*.

Table S7 Practical guidance for the use of diuretics in un patients with chronic heart failure

WHY?

To relieve breathlessness and oedema in patients with symptoms and signs of congestion.

IN WHOM AND WHEN?

(1) All patients with symptoms and signs of congestion, irrespective of LVEF.

Contraindications:

(1) Not indicated if the patient has never had symptoms or signs of congestion.
(2) Known allergic reaction/other adverse reaction (drug-specific).

Cautions/seek specialist advice:

(1) Significant hypokalaemia (K^+ ≤3.5 mmol/L)—may be made worse by diuretic.
(2) Significant renal dysfunction (eGFR <30 mL/min/1.73 m^2)—may be made worse by diuretic or patient may not respond to diuretic (especially thiazide diuretic).
(3) Symptomatic or severe asymptomatic hypotension (systolic blood pressure <90 mmHg)—may be made worse by diuretic-induced hypovolaemia.
(4) Drug interactions to look out for:
- Combination with an ACE-I, an ARB, or a renin inhibitor[a]—risk of hypotension (usually not a problem).
- Combination with other diuretics (e.g. loop plus thiazide)—risk of hypovolaemia, hypotension, hypokalaemia, and renal impairment.[b]
- NSAIDs[c]—may attenuate effect of diuretic.

WHICH DIURETIC AND WHAT DAILY DOSE?

- Loop diuretics should be first choice drug.
- Long-term use of thiazide diuretics should be avoided, as this often induces severe electrolyte disturbances that could go undetected in the ambulatory setting.

Loop diuretics:

Furosemide: starting dose 20–40 mg, usual dose 40–240 mg.

Bumetanide: starting dose 0.5–1.0 mg, usual dose 1–5 mg.

Torasemide: starting dose 5–10 mg, usual dose 10–20 mg.

Thiazides/thiazide-like diuretics:

- Bendroflumethiazide: starting dose 2.5 mg, usual dose 2.5–10 mg.
- Hydrochlorothiazide: starting dose 25 mg, usual dose 12.5–100 mg.
- Metolazone: starting dose 2.5 mg, usual dose 2.5–10 mg[b].

Non-thiazide sulfonamide:

- Indapamide: starting dose 2.5 mg, usual dose 2.5–5 mg.

WHERE?

In the community for most patients.

HOW TO USE?

- Check renal function and electrolytes.
- Start with a low dose but target an effective dose for a patient to achieve positive diuresis with a simultaneous reduction of body weight by 0.75–1.0 kg per day.
- Adjust a dose according to symptoms and/or signs of congestion, blood pressure, and renal function. Use a minimum dose necessary to maintain euvolaemia—the patient's 'dry weight' (i.e. to keep the patient free of symptoms and signs of congestion).
- Dose may need to be increased or decreased according to the patient's volume status and higher dosages are needed with lower eGFR.
- Re-check blood chemistry 1–2 weeks after an initiation and after any increase in dose (urea/BUN, creatinine, K^+, Na^+).
- When to stop uptitration, reduce dose, stop treatment—see PROBLEM SOLVING.
- Patients can be educated to alter their own diuretic dose, according to need (based on symptoms, signs, and weight changes).
- A specialist HF nurse may assist with education of the patient, follow-up (in person or by telephone), biochemical monitoring, and dose adjustment (including patient educated in dose adjustment).
- As a general rule, salt/fluid restriction is not advised. Patients should, however, be advised to avoid excessive intake.

PROBLEM SOLVING

Asymptomatic low blood pressure:

- Dose may be reduced if no symptoms or signs of congestion.

Symptomatic hypotension:

- Causing dizziness/light headedness—reduce dose if no symptoms or signs of congestion.
- Reconsider need for nitrates, calcium channel blockers,[d] and other vasodilators.

Continued

- If these measures do not solve problem, seek specialist advice.

Hypokalaemia/hypomagnesaemia:

- Increase an ACE-I/ARB dose.
- Add an MRA (or increase dose), potassium supplements; magnesium supplements.

Hyponatraemia (<135 mmol/L):

- Volume depleted:
  - Stop thiazide or switch to loop diuretic, if possible.
  - Reduce dose/stop loop diuretics if possible.
- Volume overloaded:
  - Fluid restriction should be considered.
  - Consider increasing dose of loop diuretic or consider adding acetazolamide or eGFR-based hydrochlorothiazide.
  - Consider AVP antagonist (e.g. tolvaptan if available).
  - Intravenous inotropic support.
  - Consider ultrafiltration.

Hyperuricaemia/gout:

- Consider allopurinol prophylaxis, except in case of acute gout.
- For symptomatic gout use colchicine for pain relief.
- Avoid NSAIDs.

Hypovolaemia/dehydration:

- Assess volume status; consider a diuretic dosage reduction.

Insufficient diuretic response/diuretic resistance:

- Make sure the patient does not have an excessive fluid and salt intake.
- Increase dose of diuretic.
- Consider switching from furosemide to bumetanide or torasemide.
- Add an MRA/increase dose of an MRA.
- Combine loop diuretic and thiazide/metolazone[b].
- Administer loop diuretic twice (or more times) daily or on empty stomach.
- Consider short-term i.v. infusion of loop diuretic.
- Consider ultrafiltration.

Renal impairment (rising creatinine/BUN–urea):

- Check for hypovolaemia/dehydration.
- Exclude use of other nephrotoxic agents, e.g. NSAIDs, trimethoprim.
- Withhold an MRA (if eGFR drops below 20 mL/min/1.73 m^2).
- If using concomitant loop and thiazide diuretic, stop thiazide diuretic.
- Consider reducing dose of ACE-I/ARB.
- Consider haemofiltration/dialysis.

ADVICE TO THE PATIENT

- Explain expected benefits:
  - Relieves breathlessness and oedema.
  - Symptoms improve quickly—usually within days of starting treatment.
- Advise patients to report principal adverse effects [e.g. 'fuzzy' vision changes, thirst (avoid excessive consumption of hypotonic fluids, which can cause hyponatraemia) and dizziness/symptomatic hypotension]—see PROBLEM SOLVING.
- Advise patients to avoid NSAIDs[c] not prescribed by a physician (i.e. purchased over-the-counter)—may cause diuretic resistance and renal impairment.
- Patient may be educated to adjust dose based on symptoms, signs, and changes in weight (if weighing regularly).
- Dose may need to be decreased if there is fluid loss (e.g. due to diarrhoea/vomiting, excessive sweating).

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ACE-I, angiotensin-converting enzyme inhibitor; ARB, angiotensin II receptor blocker; AVP, arginine vasopressin; BUN, blood urea nitrogen; DHF, decompensated heart failure; eGFR, estimated glomerular filtration rate; HF, heart failure; HFrEF, heart failure with reduced ejection fraction; i.v., intravenous; LVEF, left ventricular ejection fraction; MRA, mineralocorticoid receptor antagonist; NSAID, non-steroidal anti-inflammatory drug.

[a]Renin inhibitors are not recommended in HF.

[b]Usually only needed for a short period—careful monitoring of blood chemistry is essential.

[c]Avoid NSAIDs unless essential.

[d]Calcium channel blockers should be discontinued in patients with HFrEF unless absolutely necessary, and diltiazem and verapamil are potentially harmful in patients with HFrEF because of their negative inotropic action.

For further details on class of recommendation and level of evidence, see main guidelines *Section 6* as well as Supplementary data online, *Evidence Tables 29–41*. Guidance for use of diuretics in the setting of decompensated HF, see *Section 7* of the main guidelines document.

Table S8 Practical guidance for the use of mineralocorticoid receptor antagonists in patients with heart failure

WHY?
To improve symptoms, reduce the risk of HF hospitalization and increase survival.
IN WHOM AND WHEN?
(1) All patients with HF, irrespective of LVEF.
Contraindications:
(1) Known allergic reaction/other adverse reaction (drug-specific). (2) Pregnancy/risk of pregnancy and breastfeeding period.
Cautions/seek specialist advice:
(1) Significant hyperkalaemia before initiation (K^+ >5.0 mmol/L).[a] (2) Significant renal dysfunction: • Steroidal MRA: eGFR <30 mL/min/1.73 m^2.[b] • Non-steroidal MRA: eGFR <25 mL/min/1.73 m^2. (1) Drug interactions to look out for: o K^+ supplements/K^+-sparing diuretics (e.g. amiloride and triamterene; beware combination preparations with furosemide). o ACE-Is/ARBs/renin inhibitors.[b] o NSAIDs.[c] o Trimethoprim/trimethoprim-sulfamethoxazole. o 'Low-salt' substitutes with a high K^+ content. o Strong CYP3A4 inhibitors, e.g. ketoconazole, itraconazole, nefazodone, telithromycin, clarithromycin, ritonavir, and nelfinavir (when eplerenone used).
WHICH MRA AND WHAT DOSE?[d]
Eplerenone: starting dose 25 mg o.d., target dose 50 mg o.d. Spironolactone: starting dose 12.5–25 mg o.d., target dose 50 mg o.d. *Finerenone: starting and target dose is dependent on eGFR at time of initiation:* • eGFR ≤60 mL/min/1.73 m^2: starting dose 10 mg o.d., target dose 20 mg o.d. • eGFR >60 mL/min/1.73 m^2: starting dose 20 mg o.d., target dose 40 mg o.d.
WHERE?
In the community or in the hospital. Exceptions—see 'Cautions/seek specialist advice'.
HOW TO USE?
• Check renal function and electrolytes (particularly K^+). • Start with a low dose (see above[d]). • Rapid dose uptitration may be carried out in patients in hospital or who are otherwise closely monitored, tolerability permitting. In those patients where close monitoring is not possible, dose should be doubled at not less than 2-week intervals. • Aim for the target dose (see above) or, failing that, the highest tolerated dose (remember: some MRA is better than no MRA). • Recheck blood chemistry (urea/BUN, creatinine, K^+) 1–2 weeks after initiation and 1–2 weeks after final dose titration. • Monitor blood chemistry 4–6 monthly thereafter. • If K^+ rises above 5.5 mmol/L, halve a dose and monitor blood chemistry closely. • If K^+ rises to >6.0 mmol/L, interrupt MRA immediately and seek specialist advice. • If MRA has been temporarily discontinued for less than 14 days, it should, in most cases, be safe to initiate directly at the previously tolerated doses in stable patients. • If eGFR decreases to >30%, halve the dose, monitor blood chemistry closely, and consider seeking specialist advice. • A specialist HF nurse may assist with education of the patient, follow-up (in person or by telephone), biochemical monitoring, and dose uptitration.
PROBLEM SOLVING
Worsening renal function/hyperkalaemia: • See HOW TO USE? • High-normal K^+ levels may be desirable in patients with HF, especially if they are taking digoxin. • The main concern is severe hyperkalaemia; although this was uncommon in RALES, EMPHASIS-HF, and FINEARTS-HF, it has been seen more commonly in clinical practice. • It is important to avoid other K^+-retaining drugs (e.g. K^+-sparing diuretics such as amiloride and triamterene) and nephrotoxic agents (e.g. NSAIDs[c]).

Continued

- The risk of hyperkalaemia and renal dysfunction when an MRA is given to patients already taking both an ACE-I and an ARB is higher than when an MRA is added to just an ACE-I or an ARB given singly; triple combination of an ACE-I, ARB, and MRA is NOT recommended (see recommendations below).
- Some 'low-salt' substitutes have a high K^+ content.
- Male patients treated with spironolactone may uncommonly develop breast discomfort or gynaecomastia (switching to eplerenone/finerenone should be considered).

ADVICE TO THE PATIENT

- Explain expected benefits (see WHY?)
  - Treatment is given to improve symptoms, to prevent worsening of HF leading to hospital admission, and to increase survival.
  - Symptomatic improvement occurs within a few weeks to a few months of starting treatment.
  - Avoid NSAIDs[c] not prescribed by a physician (i.e. purchased over-the-counter) and salt substitutes high in K^+.
  - If diarrhoea/vomiting occurs, or there is infection with fever leading to intense sweating, patients should be made aware of the risk of dehydration and electrolyte imbalance, and they should contact the physician/nurse.

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ACE-I, angiotensin-converting enzyme inhibitor; ARB, angiotensin II receptor blocker; BUN, blood urea nitrogen; CYP3A4, cytochrome P450 isoenzyme 3A4; eGFR, estimated glomerular filtration rate; HF, heart failure; LVEF, left ventricular ejection fraction; MRA, mineralocorticoid receptor antagonist; NSAID, non-steroidal anti-inflammatory drug; o.d., once daily.
[a]It is extremely important to adhere to these cautions and doses to avoid serious hyperkalaemia.
[b]Renin inhibitors are not recommended in HF.
[c]Avoid NSAIDs unless essential.
[d]See main guidelines, Table 11.
The advice in this table represents expert opinion based upon relevant clinical trials (drugs, titration schedules, target doses, patient monitoring, treatment benefits, and reported adverse effects) and clinical experience.
For further details on class of recommendation and level of evidence, see main guidelines Section 6 as well as Supplementary data online *Evidence Tables 29–41*.

Table S9 Practical guidance for the use of sodium–glucose co-transporter 2 inhibitors in patients with chronic heart failure

WHY?

To improve quality of life, reduce the risk of HF hospitalization and increase survival.

IN WHOM AND WHEN?

(1) All patients with HF and persisting symptoms (NYHA classes II–IV), irrespective of LVEF.

Contraindications:

(1) Known allergic reaction/other adverse reaction (drug-specific).
(2) Pregnancy/risk of pregnancy and breastfeeding period.
(3) eGFR <20 mL/min/1.73 m^2.
(4) Currently not approved for type 1 diabetes due to risk of ketoacidosis.

Cautions/seek specialist advice:

(1) Symptoms of hypotension and a systolic blood pressure <95 mmHg.
(2) Glycosuria (as the consequence of SGLT2-I action) may predispose to fungal genito-urinary infections.
(3) Drug interactions to look out for:
  - Insulin, sulfonylurea derivates, and other anti-diabetic drugs predisposing to hypoglycaemia.
  - Thiazides and loop diuretics predisposing to excessive diuresis, dehydration, symptomatic hypotension, and pre-renal kidney failure.

WHAT DOSE?[a]

Dapagliflozin: starting (and target) dose 10 mg o.d.
Empagliflozin: starting (and target) dose 10 mg o.d.

WHERE?

In the community or in the hospital.

HOW TO USE?

- Check renal function when starting the therapy and monitor regularly.
- Monitor glycaemia regularly, particularly when a patient is receiving insulin. Consider modification of other diabetic drugs.
- Identify the risk factors predisposing to ketoacidosis and eliminate them if possible.
- Monitor fluid balance regularly, particularly when a patient is taking diuretics, is old, and/or frail. Consider an adjustment of diuretic therapy and fluid intake.
- A specialist HF nurse may assist with education of the patient, follow-up (in person or by telephone), and biochemical monitoring.
- If SGLT2-I has been temporarily discontinued for less than 14 days, it should, in most cases, be safe to initiate directly at the previously tolerated doses in stable patients.

Continued

PROBLEM SOLVING
Genito-urinary infections:
• Patients should be monitored in the context of symptoms and signs of genito-urinary fungal infections.
Diabetic ketoacidosis:
• Insulin predisposes to diabetic ketoacidosis.
• Euglycaemic ketoacidosis may be seen with SGLT2-I (glucose <300 mg/dL).
• Latent onset.
• Unspecific symptoms, drowsiness, and eventually coma.
Hypoglycaemia:
• Other diabetic drugs (particularly, insulin, and/or sulfonylurea derivatives) may predispose to hypoglycaemia; in this case, the diabetic treatment strategy needs to be modified.
• Dehydration, hypotension, and prerenal kidney failure.
• SGLT2-I may intensify diuresis, particularly when accompanied by sacubitril–valsartan and diuretic therapy.
• Diuretic doses along with fluid intake should be balanced in order to avoid dehydration, symptomatic hypotension, and pre-renal kidney failure.
• Elderly and frail patients are at particular risk of developing these complications.
Worsening renal function:
• eGFR drops <20 mL/min/1.73 m^2: seek other causes. Continuation of SGLT2-I is advised but may be after close monitoring.
ADVICE TO THE PATIENT
• Explain expected benefits (see WHY?):
○ Treatment is given to improve quality of life, to prevent worsening of HF leading to hospital admission, and to increase survival (to reduce the risk of cardiovascular and all-cause death).
○ Improvement in quality of life occurs within a few weeks to a few months of starting treatment.
• Due to dapagliflozin action, glycosuria is an expected finding on urinalysis.
• Patients should be made aware of the risk of dehydration, hypotension, hypoglycaemia, ketoacidosis, and fungal genito-urinary infections, and in these cases they should contact the physician/nurse.

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eGFR, estimated glomerular filtration rate; HF, heart failure; LVEF, left ventricular ejection fraction; NYHA, New York Heart Association; o.d., once daily; SGLT2-I, sodium–glucose co-transporter 2 inhibitor.
[a]See main guidelines, Table 11.
The advice in this table represents expert opinion based upon relevant clinical trials (drugs, titration schedules, target doses, patient monitoring, treatment benefits, and reported adverse effects) and clinical experience.
For further details on class of recommendation and level of evidence, see main guidelines *Section 6* as well as Supplementary data online, *Evidence Tables 29–41*.

3.2. Guideline-directed interventional therapies for chronic heart failure

3.2.1. Implantable cardioverter defibrillator for primary prevention of sudden cardiac death

The SCD-HeFT study showed that, compared with amiodarone, implantable cardioverter-defibrillator (ICD) therapy was associated with a 23% reduction in total mortality in patients with symptomatic HFrEF (NYHA class II–III).[13] In the MADIT-II trial, 1232 post-myocardial infarction (MI) patients were randomized to ICD or conventional medical therapy alone. ICD therapy was associated with a 31% reduction in total mortality.[14] In both trials, patients were recruited from 1997 to 2001. The majority of patients were on foundational medical therapy (FMT), which at the time only included beta-blockers and renin–angiotensin–aldosterone system inhibition. Since then, management of HFrEF has evolved considerably, which may have altered the background risk of sudden death in these patients,[15,16] although not to the extent that ICDs are no longer necessary.[15]

The overall use of guideline-directed interventional therapy is low; however, with significant regional variations. In the PARADIGM-HF trial, for example, only 14.8% of patients received an ICD, despite 85% being eligible.[17] A propensity score-adjusted post-hoc analysis, on the background of FMT, examined the effect of angiotensin receptor–neprilysin inhibitors (ARNIs) on sudden cardiac death (SCD). Although ARNIs reduced SCD risk in patients without an ICD [hazard ratio (HR) 0.81; 95% confidence interval (CI) 0.67 to 0.98], a greater reduction (HR 0.49; 95% CI 0.25 to 0.99) was observed in those with an ICD.[18] These findings suggest that FMT does not totally eliminate the risk of SCD and that the best outcomes are achieved when FMT and ICD are used in combination.

3.2.2. Non-transvenous implantable cardioverter-defibrillator and wearable cardioverter-defibrillator

The subcutaneous ICD consists of a generator implanted in a subfascial or submuscular pocket and a subfascial lead over the sternum.

In the HF-OPT study, which explored the use of wearable cardioverter-defibrillators (WCD) during uptitration of FMT in patients with newly diagnosed HFrEF and an LVEF ≤35%, 46% of patients reached LVEF >35% within 90 days.[19] Sustained ventricular tachyarrhythmias were observed in 24 WCD carriers (1.8%) within 90 days of initiation of FMT. Although this observational study explored the proof-of-principle of 'protected' uptitration, it was not blinded or randomized. In addition, patient tolerability of WCDs is questionable. The VEST trial failed to show a reduction in arrhythmic death with WCDs in patients with a LVEF ≤35% following an MI.[20]

3.2.3. Cardiac resynchronization therapy

Table S10 Landmark trials with cardiac resynchronization therapy

Year	Study[a]	Number	Design	Inclusion criteria	Comparison	Effect of CRT
2001	MUSTIC-SR[21]	67	Single-blind, crossover RCT	NYHA III, LVEF <35%, QRS ≥150 ms, LVEDd >60 mm, 6MWD <450 m	CRT vs VVI (no pacing indications)	CRT improved QoL, walking distance, peak VO_2; reduced hospitalizations
2002	MUSTIC-AF[22]	43	Single-blind, crossover RCT	NYHA III, LVEF <35%, RV-paced QRS ≥200 ms, LVEDd >60 mm, 6MWD <450 m	VVIR vs BiV	CRT improved 6MWD, peak VO_2, QoL, and NYHA class; reduced hospitalizations (but no difference on intention-to-treat analysis)
	PATH-CHF[23]	42	Single-blind, crossover RCT	NYHA II–IV, LVEF <35%, PR-interval ≥150 ms, QRS ≥120 ms	RV vs LV vs BiV	CRT improved NYHA class, QoL, and walking distance
	MIRACLE[24]	453	Double-blind RCT	NYHA III–IV, LVEF <35%, QRS ≥130 ms, LVEDd >55 mm	CRT-on vs CRT-off	CRT improved NYHA class, QoL, walking distance, LVEF, peak VO_2, mitral regurgitation; reduced hospitalizations
2003	MIRACLE-ICD[25]	369	Double-blind RCT	NYHA III–IV, LVEF <35%, QRS ≥130 ms, LVEDd >55 mm	CRT-D vs ICD	CRT improved NYHA class, QoL, and walking distance, and reduced hospitalizations
	CONTAK-CD[26]	490	Double-blind, crossover RCT	NYHA II–IV, LVEF <35%, QRS ≥120 ms, ICD indication	CRT-on vs CRT-off	CRT improved peak VO_2 and walking distance, not NYHA or QoL; reduced LV volumes and improved LVEF; no effect on HF progression
2004	COMPANION[27]	1520	Unblinded RCT (1:2:2)	NYHA III–IV, LVEF <35%, QRS >120 ms	OMT vs CRT-P or CRT-D	CRT-D and CRT-P reduced composite of all-cause mortality and hospitalization
	MIRACLE-ICD II[28]	186	Double-blind RCT	NYHA II, LVEF ≤35%, QRS ≥130 ms, ICD indication	CRT-on vs CRT-off	CRT reduced LV volumes and LVEF and improved CCS; no effect on QoL, walking distance, or peak VO_2
2005	CARE-HF[29]	813	Unblinded RCT	NYHA III–IV, LVEF <35%, QRS >120 ms	CRT-P vs OPT	CRT reduced total mortality and HF hospitalizations
2006	HOBIPACE[30]	30	Double-blind, crossover RCT	Pacing indications, LVEF <40%, LVEDd >60 mm	CRT-P vs RV pacing	CRT reduced LV volumes and improved QoL, LVEF, peak VO_2
2007	RethinQ[31]	172	Double-blind RCT	NYHA III, LVEF <35%, QRS <130 ms, echo dyssynchrony	CRT-on vs CRT-off in CRT-D recipient	CRT improved NYHA class, but not walking distance, LVEF, or QoL
2008	PROSPECT[32]	498	Prospective, observational	NYHA II–IV, LVEF <35%, QRS >130 ms, OMT	Echo dyssynchrony measures as predictor of CCS and LVRR	Echo dyssynchrony measures did not predict outcome after CRT
	REVERSE[33]	610	Double-blind RCT (2:1)	NYHA I–II, LVEF <40%, QRS >120 ms	CRT-on vs CRT-off (ICD on)	CRT reduced HF hospitalization and improved LVEF and NYHA class; no effect on mortality

Continued

2009	MADIT-CRT[34]	1820	Single-blind RCT	NYHA I–II, LVEF <30%, QRS >130 ms	CRT-D vs ICD	CRT-D reduced HF events; no effect on mortality
2010	RAFT[35]	1798	Double-blind RCT	NYHA II–III, LVEF <30%, QRS >120 ms	CRT-D vs ICD	CRT reduced total mortality and HF hospitalizations
2013	BLOCK-HF[36]	691	Single-blind RCT	NYHA I–III, AV block, LVEF <50%	CRT vs RV pacing	CRT reduced composite of total mortality, HF event or 15% increase in LVESVi
	Echo-CRT[37]	809	Double-blind RCT	NYHA III–IV, LVEF <35%, QRS <130 ms, echo dyssynchrony	CRT-on vs CRT-off	No effect on composite of total mortality or HF hospitalization; higher total mortality with CRT-on
	LESSER EARTH[38]	85	Double-blind RCT	NYHA III–IV, LVEF ≤35%, QRS <120 ms	CRT-D vs ICD	Stopped prematurely after recruiting 85 patients: CRT reduced walking distance and increased QRS duration; trend towards increased HF hospitalizations
2023	BUDAPEST-CRT[39]	360	Unblinded RCT (3:2)	NYHA II–IV, LVEF ≤35%, ≥20% RV pacing, paced QRS ≥150 ms	CRT-D or ICD	Upgrade to CRT-D reduced the composite endpoint of total mortality, HF hospitalization or a <15% reduction in LVESV (odds ratio 0.11; $P < .001$); total mortality and HF hospitalization also reduced (HR 0.27, $P < .001$)

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6MWD, 6-min walk distance; AV, atrioventricular; BiV, biventricular; CCS, clinical composite score; CRT, cardiac resynchronization therapy; CRT-D, cardiac resynchronization therapy with defibrillator; CRT-P, cardiac resynchronization with pacemaker; HF, heart failure; HR, hazard ratio; ICD, implantable cardiac defibrillator; LV, left ventricular; LVEDd, left ventricular end-diastolic diameter; LVEF, left ventricular ejection fraction; LVESV, left ventricular end-systolic volume; LVESVi, left ventricular end-systolic volume index; LVRR, left ventricular reverse remodelling; NYHA, New York Heart Association; OMT, optimal medical therapy; OPT, optimal pharmacological therapy; QoL, quality of life; RCT, randomized clinical trial; RV, right ventricular; VO_2, oxygen consumption; VVI, ventricular–ventricular interaction; VVIR, ventricular–ventricular interaction rate responsive.
[a]See Supplementary data online, *Table S1* for full names of trials.

3.2.4. Conduction system pacing

Direct stimulation of the His bundle or left bundle branch is termed 'selective', whereas indirect stimulation involving the additional recruitment of the local myocardium is termed 'non-selective'.

Although conduction system pacing is, in essence, a form of cardiac resynchronization therapy (CRT), the term CRT will be reserved for trans-coronary vein (trans-CV) CRT in this guideline.

In an RCT of 50 patients with HFrEF, changes in LVEF at 6 months were similar to trans-CV CRT.[40] Despite these findings, low implantation success rates (≤75%) and long-term complications, notably the development of atrioventricular block (as high as 19.6%),[41] unacceptable thresholds, and loss of capture at follow-up[42,43] have led experienced operators to largely abandon His bundle pacing.

In a prospective study of 187 patients with an LVEF ≤40%, the implant success rate of left bundle branch area pacing (LBBAP) was 81% in patients with a left bundle branch block (LBBB). However, this figure was only 54.5% in those with a non-LBBB.[44] In the only RCT comparing LBBAP with trans-CV CRT, a higher LVEF was observed after LBBAP.[45] Although it is reasonable to use LBBAP if implantation of a trans-CV lead is not successful, LBBAP cannot be recommended as first line in CRT candidates.[46] In this context, no studies have satisfactorily addressed whether LBBAP should be used in CRT 'non-responders'.

4. Decompensated heart failure

4.1. Diagnosis

Table S11 Admission investigations in decompensated heart failure

Admission diagnostic tests	Role
Pulse oximetry	To guide management
Full blood count	Identifies trigger, informs prognosis
Urea, creatinine, and electrolytes	To guide management, informs prognosis
Natriuretic peptides	Diagnosis, informs prognosis
Troponin	Identifies trigger, informs prognosis
TSH	Identifies trigger
D-dimer	Differential diagnosis (if PE is suspected or cannot be excluded)
Pro-calcitonin/CRP	Differential diagnosis (if sepsis is suspected), identifies trigger
Lactate	Diagnosis (especially if cardiogenic shock is suspected), to guide management, informs prognosis
Urine spot sodium	To guide the management of congestion, informs prognosis
Chest X-ray	Diagnosis and identifies trigger
ECG	Aetiology and identifies trigger
Echocardiogram	Diagnosis, aetiology
LUS	Diagnosis (when trained personnel is available)

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CRP, C-reactive protein; ECG, electrocardiogram; LUS, lung ultrasound; PE, pulmonary embolism; TSH, thyroid-stimulating hormone.

4.2. General management

4.2.1. Phase 1 (initial management)

The Emergency Heart Failure Mortality Risk Grade for 30-day mortality (EHMRG30-ST) is a composite of two different risk models: the 7-day risk score and a 30-day probability of death.[47] All values for the 7-day and 30-day risk scores are obtained from values taken in the emergency department or recorded by the emergency medical services.

Table S12 Tests for stabilized, ward-based patients during decongestion and medical optimization

Monitoring tests	Frequency	Role
Clinical		
Symptom assessment (including congestion/ perfusion), NYHA class	As clinically needed, daily[a]	To assess response, to guide treatment
Basic vitals (including BP, HR, diuresis, fluid balance)	As clinically needed, daily[a]	To assess safety, response, to guide treatment
Clinical examination of fluid status, diuretic response	Daily[a]	To assess response, to guide treatment
Weight recording	Daily	To assess response, guide treatment
Laboratory		
Urea, creatinine, electrolytes	Daily[a]	To guide treatment and for early detection of complications
Urine spot sodium	Daily[a] until congestion persists	To assess response, guide treatment
NT-proBNP	Pre-discharge	To guide management in the early post-discharge phase, to inform prognosis
Iron status	Once	To assess the requirement for treatment
Imaging		
Focused echocardiogram for markers of congestion (LV filling pressures, TAPSE, sPAP, IVC)	As needed, until stabilization where available, pre-discharge	To assess response and assess for parameters of congestion
LUS	Pre-discharge	To guide management and inform prognosis

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BP, blood pressure; HR, heart rate; IVC, inferior vena cava; LUS, lung ultrasound; LV, left ventricular; NT-proBNP, N-terminal pro-B-type natriuretic peptide; NYHA, New York Heart Association; sPAP, systolic pulmonary artery pressure; TAPSE, tricuspid annular plane systolic excursion.
[a]Recommended frequency is once a day at a minimum.

Table S13 Pharmacotherapy and practical guidance for the use of diuretics in decompensated heart failure

	Acetazolamide	Loop diuretics	Thiazide-like diuretics	MRAs (spironolactone, eplerenone, etc.)[a]	Amiloride
Site of action	Proximal nephron	Ascending loop of Henle	Early distal convoluted tubule	Late distal tubule	Late distal tubule
Starting dose/ usual chronic dose	Oral: 250–375 mg IV: 500 mg	**Furosemide:** 20–40/ 40–240 mg[b] **Bumetanide:** 0.5–1.0/ 1–5 mg[b] **Torsemide:** 5–10/10–20 mg[b]	**HCTZ:** 25/12.5–100 mg[c] (eGFR >50 mL/min:25 mg daily; eGFR 20–50 mL/min: 50 mg daily; and eGFR <20 mL/min: 100 mg daily) **Metolazone:** 2.5/2.5–10 mg[c] **Chlorthalidone:** 25/ 25–200 mg[c] **Chlorothiazide:** 500–1000 mg (IV formulation available)	**Spironolactone:** 25/ 25–50 mg **Eplerenone:** 25/25–50 mg **K^+ Canrenoate:** 25–200 mg (not for chronic use)	5/10 mg
Maximum recommended total daily dose	Oral: 500 mg 3×/ day IV: 500 mg 3×/day	**Furosemide:** 600 mg (up to 1000 mg in patients with severe CKD) **Bumetanide:** 10–15 mg **Torsemide:** 200–300 mg	**HCTZ:** 200 mg **Metolazone:** 20 mg **Chlorthalidone:** 100 mg **Chlorothiazide:** 1000 mg	50–100 mg (up to 400 mg used in hepatology)	20 mg
Half-life	2.4–5.4 h	**Furosemide:** 1.5–3 h **Bumetanide:** 1–1.5 h **Torsemide:** 3–6 h	**HCTZ:** 6–15 h **Metolazone:** 6–20 h **Chlorthalidone:** 45–60 h	**Canrenone:** 16.5 h[d] **Eplerenone:** 3–6 h	Normal GFR: 6–9 h GFR <50 mL/min: 21–144 h
Onset	PO: 1 h IV: 15–60 min	PO: 0.5–1 h[e] IV: 5–10 min[e] SC: 0.5 h[e]	PO: 1–2.5 h IV (Chlorothiazide is IV available, onset action): 30 min	PO: 48–72 h[d] IV: Potassium Canrenoate: 2.5 h	PO: 2 h IV: Not available
Oral bioavailability	Dose-dependent (dose >10 mg/kg exhibits variable uptake)	**Furosemide:** 10%–100% **Bumetanide:** 80%–100% **Torsemide:** 80%–100%	**HCTZ:** 65%–75% **Metolazone:** 60%–65%[f] **Chlorthalidone:** Unknown **Chlorothiazide:** 9%–56%	**Spironolactone:** ~90% **Eplerenone:** 69%	30%–90%
Enteral absorption affected by food	May be taken with food. Food decreases symptoms of GI upset	**Furosemide and Bumetanide:** yes (absorption slowed) **Torsemide:** not affected	Unknown	**Spironolactone:** ↑ with high-fat food **Eplerenone:** Unknown	Unknown
Potency (FENa %)[g]	4%	20%–25%[e]	5%–8%	2%	2%
Additional notes	Studies suggest that acetazolamide is the medication of choice to combine	Loop diuretics represent the first choice among diuretics	More frequently associated with worsened renal function and	Medication of choice to combine with loop diuretics in patients with hypokalaemia	

Continued

	with loop diuretics in patients with metabolic alkalosis (bicarbonate levels ≥27 mmol/L)		hypokalaemia. Avoid if hypokalaemia and hyponatraemia		

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CKD, chronic kidney disease; (e)GFR, (estimated) glomerular filtration rate; FENa, fractional excretion of sodium; GI, gastro-intestinal; HCTZ, hydrochlorothiazide; HF, heart failure; IV, intravenous; MRA, mineralocorticoid receptor antagonist; PO, per os; SC, subcutaneous.
Diuretic agents are reflected from the site of action; from proximal nephron to distal nephron.
[a]Minimal diuretic effect.
[b]Dose of intravenous and oral loop diuretics are similar.
[c]Only PO use in decompensated HF, thiazides are not recommended for daily ambulatory use in chronic stable HF.
[d]Canrenone is the active metabolite of spironolactone. Intravenous potassium canrenoate is the IV formulation and is metabolized to canrenone, resulting in significant plasma levels after 2.5 h of administration.
[e]Generally similar for different loop diuretics.
[f]Variations between pharmaceutical brands of metolazone exist.
[g]Tested in non-HF patients. FENa is the percentage of the sodium filtered by the kidney, which is ultimately excreted in the urine. It is measured based on plasma and urinary sodium. In clinical use, FENa can be calculated as part of the evaluation of diuretic effectiveness. The normal value depends primarily on the GFR of the patient but is commonly <2% in patients with relatively intact renal function. Diuretic agents increase FeNa with loop diuretic agents to be the most potent ones. FENa = 100 × (sodium urine × creatinine plasma)/(sodium plasma × creatinine urine).

4.3. Vasodilators

Intravenous (i.v.) vasodilators, through venous and arterial vessel dilatation, lead to a reduction in venous return to the heart, less congestion, lower afterload, increased stroke volume, and consequent relief of symptoms.

Because of their mechanisms of action, i.v. vasodilators may be more effective in patients with pulmonary congestion caused by an increase in afterload and fluid redistribution to the lungs in the absence or with minimal fluid accumulation.

The most frequently used i.v. vasodilators are nitrates and sodium nitroprusside, (*Table S14*). Nitrates act mainly on peripheral veins whereas nitroprusside is more a balanced arterial and venous dilator.

They may be started at low doses and uptitrated to achieve clinical improvement and blood pressure control. Nitrates are generally administered with an initial bolus followed by continuous infusion. However, they may also be given as repeated boluses. Nitroglycerine can be given as 1–2 mg boluses in severely hypertensive patients with acute pulmonary oedema.

Care should be taken to avoid hypotension due to an excessive decrease in preload and afterload. For this reason, they should be used with extreme caution in patients with left ventricular hypertrophy and/or severe aortic stenosis. However, favourable effects were described in patients with left ventricular systolic dysfunction and aortic stenosis when vasodilators were given with careful monitoring of haemodynamic parameters.

4.4. Inotropes/inodilators and vasopressors

Table S14 Intravenous vasodilators for decompensated heart failure

Vasodilator	Mechanism of action	Vascular bed target	Dosing	Main side effects	Other
Sodium nitroprusside	NO donor	Balanced arterial and venous dilator	Start with 0.3 μg/kg/min and increase up to 5 μg/kg/min	Hypotension, isocyanate toxicity	Light sensitivity
Nitroglycerine	NO donor	Predominantly peripheral veins + large arteries	Start with 10–20 μg/min, increase up to 200 μg/min (consider starting with 1–2 mg boluses in severely hypertensive patients with acute pulmonary oedema)	Hypotension, headache	Tolerance in continuous use

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NO, nitric oxide.

Table S15 Pharmacology of calcium sensitizers (levosimendan) and phosphodiesterase III inhibitors (milrinone and enoximone)

Drug	Receptor targets	Mechanism of action	Main clinical effects	Bolus	Infusion rate	Elimination $t_{1/2}$	Other considerations
Levosimendan	• Calcium sensitization: binds to troponin C → enhances actin–myosin interaction without raising intracellular Ca^{2+} • ATP-dependent K^+ channel opening: vasodilatation (systemic, pulmonary, coronary) • PDE3 inhibition: ↑ cAMP	• Increased contractility without increased oxygen demand • Coronary, pulmonary, and systemic vasodilatation	• ↓ Pulmonary capillary wedge pressure • ↑ Cardiac output (vs placebo or dobutamine)	Not recommended—associated with initial hypotension/arrhythmias	Standard: 0.1 μg/kg/min Adjust to 0.05 or 0.2 μg/kg/min based on BP and response	• Levosimendan $t_{1/2}$ ≈ 1 h • Active metabolite (OR-1896) $t_{1/2}$: 70–90 h → effects can last weeks	Potential adverse effects: hypotension (especially with bolus); risk of arrhythmia (less than dobutamine). Unique feature: long-lasting effect through long-acting active metabolite, allowing periodic administration. Used in acute decompensated heart failure but also considered for intermittent therapy due to long-acting metabolite
Milrinone	PDE3inhibition: ↑ cAMP	• Inhibits PDE3 → increases intracellular cAMP → enhances cardiac contractility (↑ inotropy) and vasodilatation • Positive lusitropy (↑ myocardial relaxation)	• ↑ Cardiac output • ↓ Filling pressures • ↓ Pulmonary arterial pressures		0.375–0.75 μg/kg/min	Plasma $t_{1/2}$: 2–2.5 h	
Enoximone	PDE3inhibition: ↑ cAMP	Inhibits PDE3 → increases intracellular cAMP → enhances cardiac contractility (↑ inotropy) and vasodilatation	• ↑ Cardiac output • ↓ Filling pressures • ↓ Pulmonary arterial pressures		5–20 μg/kg/min	~4 h	

ATP, adenosine triphosphate; BP, blood pressure; cAMP, cyclic adenosine monophosphate; PDE3, phosphodiesterase III; $t_{1/2}$, half-life.

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Table S16 Pharmacology of beta-adrenergic drugs (dobutamine, epinephrine, norepinephrine, dopamine)

Drug	Receptor targets (+)	Mechanism of action	Main clinical effects	Bolus	Infusion rate	Elimination $t_{1/2}$	Special considerations
Dobutamine	β_1 (β_2)	β_1 stimulation → ↑ cAMP → ↑ Ca^{2+} influx → ↑ inotropy and chronotropy	↑ Cardiac output, ↓ LV filling pressure, possible ↓ BP due to β_2 vasodilatation	–	2–20 μg/kg/min	±2 min	Effects may be blunted by β-blockers; reflex vasodilatation may lower BP
Epinephrine	α_1, β_1, β_2	Low dose: β_1 → ↑ inotropy and chronotropy High dose: α_1 → vasoconstriction	Potent ↑ CO and HR; at higher doses, ↑ SVR	+	0.05–0.5 μg/kg/min	±2.5 min	High arrhythmogenic potential; used mainly in rescue or cardiac arrest situations
Norepinephrine	α_1 (β_1)	Strong α_1 stimulation → vasoconstriction; β_1 → mild inotropy	↑ SVR, modest ↑ CO; used to restore BP in hypotensive patients	–	0.2–1.0 μg/kg/min	±1–2 min	Preferred vasopressor in shock; less effective as a pure inotrope; rarely used in chronic HF
Dopamine	Low dose: dopamine receptors Moderate dose: β_1 High dose: α_1	Low dose (<3 μg/kg/min): renal vasodilatation Moderate dose (3–5 μg/kg/min): ↑ inotropy (β_1) High dose (>5 μg/kg/min): vasoconstriction (α_1)	Selective renal vasodilatation at low dose; ↑ CO at moderate dose; vasoconstriction at high dose	–	Low: <3 μg/kg/min Moderate: 3–5 μg/kg/min High: >5 μg/kg/min	±2 min	Not commonly used in HF due to risk of tachyarrhythmias and unpredictable BP response

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BP, blood pressure; cAMP, cyclic adenosine monophosphate; CO, cardiac output; HF, heart failure; HR, heart rate; LV, left ventricular; SVR, systemic vascular resistance; $t_{1/2}$, half-life.

5. Advanced heart failure

5.1. Epidemiology, diagnosis, and prognosis

Table S17 Definition of malnutrition, cardiac cachexia, and sarcopenia

Disorder	Definition	Prevalence/impact in HF	Management considerations
Malnutrition	Deficiency due to poor intake or absorption of nutrients (usually the latter in patients with HF). It leads to a decrease fat-free mass, ↓ physical/mental function. Malnutrition can be assessed according to scores, like the prognostic nutritional index [PNI: albumin (g/L) + 5×total lymphocyte count ×10^9/L)], controlling nutritional status (CONUT: sum of albumin, cholesterol and lymphocyte count), and the geriatric nutritional risk index [GNRI: (1.489 × albumin (g/L)) + 41.7 × (weight/ideal weight)].	Common in HF, especially advanced stages. Impact on prognosis.	Nutritional support may be useful.
Cachexia	Multifactorial syndrome with irreversible weight/muscle/fat loss. It can be due to ↑ protein catabolism, not corrected by nutrition alone. Severely ↓ physical/mental function.	50% 18-month mortality vs 17% in non-cachectic HF patients.	Considered an advanced form of disease-related malnutrition; needs comprehensive treatment.

Continued

	It can be defined as weight loss >7.5% in the absence of oedema.		
Sarcopenia	Progressive muscle loss and strength decline. It affects muscle mass, strength, and physical performance.	20× more prevalent in HF with reduced LVEF than in healthy peers.	Focus on resistance training, nutritional support, and treating underlying HF.

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HF, heart failure; LVEF, left ventricular ejection fraction.

5.2. Management

Table S18 Definition of inotrope dependency

FAILURE to wean intravenous inotropic support[a] within 72 h[b]
WITHOUT: (i) Development of symptomatic arterial hypotension OR (ii) Worsening renal or hepatic function defined as eGFR decrease >30%, or clinically important elevation in liver enzymes or INR OR (iii) Worsening congestion leading to or upholding NYHA class IV symptoms[c]
The diagnosis of inotrope dependence should not be made during simultaneous introduction or uptitration of beta-blockers or RAS inhibitors

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eGFR, estimated glomerular filtration rate; INR, international normalized ratio; NYHA, New York Heart Association; RAS, renin–angiotensin system.
[a]Epinephrine, norepinephrine, dopamine, dobutamine, milrinone.
[b]If the patient has received continuous intravenous inotropic support >7 days, inotrope dependence is defined as failure to reduce infusion rate at 72 h after each attempt to reduce. In the case of levosimendan infusion, dependence is defined as need for new infusion <10 days after the former.
[c]In the absence of reduction in loop diuretic dose.

5.2.1. Pharmacological therapy and renal replacement

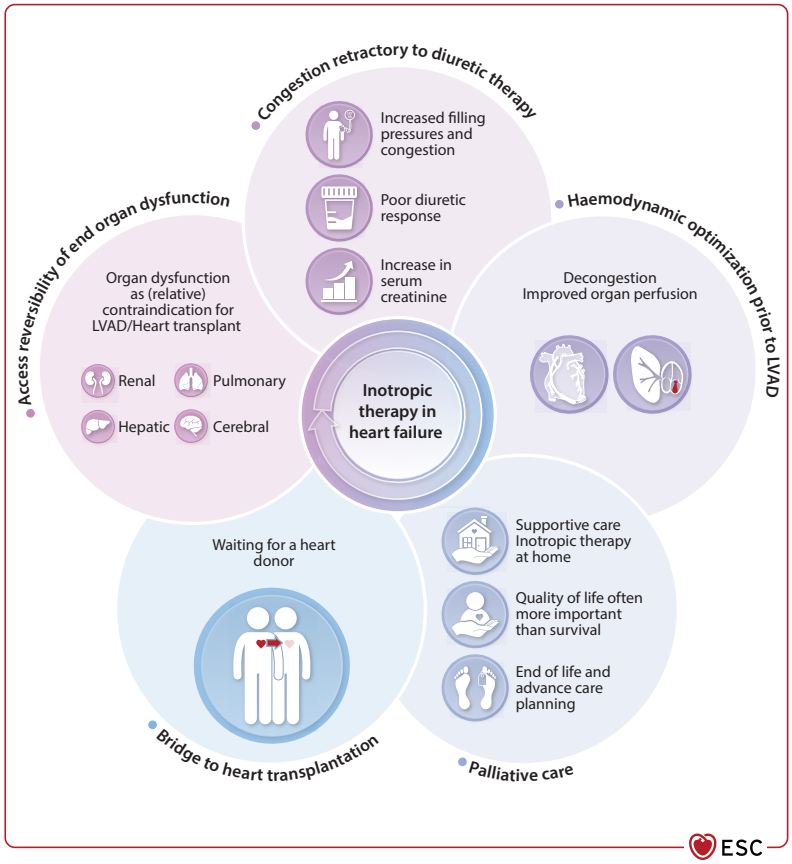


Figure S2 Applications of inotropes in advanced heart failure. LVAD, left ventricular assist device. Adapted from Gustafsson *et al.*[48]

5.2.2. Mechanical circulatory support

Table S19 Terms describing various indications for mechanical circulatory support

Bridge to decision/bridge to bridge	Use of temporary MCS in patients with cardiogenic shock until haemodynamic parameters and end-organ perfusion are stabilized, contraindications for durable MCS are excluded, and additional therapeutic options, including durable MCS therapy or heart transplantation, can be evaluated
Bridge to surgery	Use of temporary MCS in patients with cardiogenic shock with an indication to surgery as a bridge to surgery
Bridge to candidacy	Use of durable and temporary MCS (usually LVAD) to improve end-organ function and/or to make an ineligible patient eligible for heart transplantation
Bridge to transplantation	Use of durable and temporary MCS to keep a patient alive who is otherwise at high risk of death before transplantation until a donor organ becomes available
Bridge to recovery	Use of MCS to keep a patient alive until cardiac function recovers sufficiently to remove MCS
Destination therapy	Use of durable MCS as an alternative to transplantation in patients with end-stage HF ineligible for transplantation

HF, heart failure; LVAD, left ventricular assist device; MCS, mechanical circulatory support.

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5.2.3. Symptom control and end-of-life care

Table S20 Core components of end-of-life care in advanced HF

Recognition of EOL in HF
• Frequent hospitalizations or emergency visits due to decompensated HF. • Increasing ICD shocks. • Progressive functional decline (physical and mental), dependence in most activities of daily living. • Progressive decline in quality of life. • Heart transplantation and durable MCS ruled out. • Refractory symptoms despite optimal medical and device management. • Signs of advanced HF: end-organ dysfunction, cardiac cachexia, frailty, hypotension. • Clinically judged to be close to EOL.
Communication and decision-making
Communication: • Early and ongoing conversations. • Initiate advance care planning early in the HF trajectory. • Discuss the difference between palliative care and EOL care and the appropriate timing for integrating each into HF management. • Engage in sensitive, patient-centred discussions about prognosis (including uncertainty), treatment goals, and expectations, ensuring patients and families understand their options in the context of HF. • Discuss advanced directives, resuscitation preferences, and goals of care. • Discuss the option of deactivating ICDs to prevent unnecessary shocks. • Identify preferred location of death and facilitate where possible. • Explain the dying process and consider the impact this has on decision-making. Shared decision-making: • Involve patients, caregivers, and wider HF-MDT in discussions about realistic treatment goals, care available, and location of care. • Review patient and caregiver expectations and priorities for quality and quantity of life.

Continued

Ongoing assessment and management of symptoms
Assessement: • Implement a management plan that facilitates monitoring and reassessment of the patient's symptoms and palliative care needs over time, adjusting the care plan in response to changes in condition or treatment goals. • Evaluate social determinants such as socioeconomic status, education, language barriers, and healthcare accessibility, and their impact on EOL care and outcomes, tailoring care to address these challenges. *Pharmacological management:* • Optimize diuretics for fluid management and make an individualized approach to alteration to HF FMT. • Assess polypharmacy and adjust therapies to those alleviating symptoms or preventing distress. • Manage symptoms including nausea, anxiety, pain, and depression. *Non-pharmacological:* • Use oxygen therapy if indicated for symptom relief. • Provide advice on position, use of fans and breathing techniques. • Relax previously introduced fluid and sodium restrictions in appropriate patients. • Provide dietary advice to manage nausea and vomiting. • Consider holistic and complimentary therapies available through local hospice.
Multidisciplinary care
• Integrate palliative care teams in HF EOL decision-making. • Collaborate with palliative care teams, social workers, benefits agencies, and spiritual care providers to develop comprehensive, patient-centred care plans that address physical, financial, emotional, and spiritual needs. • Consider hospice referral when shifting to comfort-orientated care with physical, emotional, and spiritual support.
Ethical considerations
• Respect patient autonomy and involve them in decisions relating to their care. • Follow local ethical and legal frameworks to support decisions relating to withdrawal of treatments and assisted dying.
Support for families and caregivers
• Include families and caregivers in discussions about EOL with consent from the patient • Recognize psychosocial, spiritual, and emotional distress and signpost to appropriate support.
Skills and attitudes
• Respect and advocate for the patient's choice including treatment withdrawal and place of death, even if it differs from the family's or healthcare team's preferences. • Recognize own limitations in the assessment and management of the patient with HF and palliative or EOL needs and utilize the expertise in the MDT. • Consider the cultural and ethical beliefs, values, and traditions relating to symptoms management, EOL care, and death, ensuring care plans are respectful and aligns with the patient's cultural perspectives. • Recognize the impact of providing EOL care to the healthcare professionals involved and facilitate opportunities to debrief including case reviews.

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EOL, end of life; FMT, foundational medical therapy; HF, heart failure; ICD, implantable cardioverter-defibrillator; MCS, mechanical circulatory support; MDT, multidisciplinary team.

Treatment for symptoms needs to be considered in end of life and might include additional interventions on top of optimal medical therapy:

- Breathlessness: repeat doses of opioids may be considered for the relief of dyspnoea; however, their effectiveness is not demonstrated. While using opioids, all patients should be guided about opioid side effects such as constipation and nausea, urinary retention, and mental status changes. Benzodiazepines may be considered as a second- or third-line treatment when opioids and non-pharmacological measures have failed to control breathlessness. Increasing the inspired oxygen concentration may provide relief of dyspnoea.
- Pain: non-pharmacological management can be helpful. In addition, opioids, oxycodone, hydromorphone, and fentanyl are generally viewed as safe options and can be provided orally, intravenously, and transdermally.
- Anxiety and depression: adequate conventional treatment should be offered.[49]

6. Cardiovascular comorbidities

6.1. Arrhythmias and conduction disturbances

6.1.1. Atrial fibrillation

In patients with chronic HF, the role of rhythm control by means of catheter ablation is debated. The CASTLE-AF trial enrolled patients with HFrEF and an ICD, who did not respond to, or had unacceptable side effects from, or were unwilling to take, antiarrhythmic drugs. In CASTLE-AF there was a statistically significant reduction in morbidity and mortality in patients receiving catheter ablation, but with a wide degree of uncertainty around the estimate.[50] In the CASTLE-HTx trial, patients referred for heart transplantation evaluation were enrolled. The trial was stopped early as catheter ablation combined with medical therapy significantly reduced the composite of death from any cause, implantation of left ventricular assist device, or urgent heart transplantation compared with medical treatment in this population.[51] Important limitations of CASTLE-AF and CASTLE-HTx have to be taken into account, including the highly selected populations enrolled, the unblinded design, and the low number of events.

The RAFT-AF study randomized patients with HFrEF or HFpEF to ablation or standard treatment.[52] The trial was stopped early due to apparent futility. There was an improvement in all-cause mortality or HF events in favour of ablation; however, the results did not meet statistical significance. A greater improvement in QoL was observed in the intervention vs control arm.

The CABANA trial was not an HF trial, but 35% of participants were reported as having HF.[53] The overall trial randomized 2204 patients with symptomatic AF to catheter ablation or medical therapy and showed no difference between the two treatment strategies on the composite endpoint of death, disabling stroke, serious bleeding, and cardiac arrest. There was a significant improvement in patients' symptoms with catheter ablation. In a post-hoc analysis of the 778 patients reported to have HF (NYHA class ≥II), catheter ablation reduced the risk of the primary composite endpoint, as compared with medical treatment (HR 0.64; 95% CI 0.41 to 0.99).[54] There has been debate whether these 778 patients constitute a representative HF population, as >90% had LVEF >35% and the rate of HFH was only 9% at 5 years.

The AMICA trial included 140 patients with persistent/long-standing persistent AF and LVEF ≤35% who were randomly allocated to catheter ablation of AF or optimized medical therapy.[55] The study was prematurely interrupted for futility because the primary endpoint (LVEF increase) did not differ in the two treatment groups.[55]

In the CAMTAF trial, 50 patients with HFrEF and persistent AF were randomly assigned to catheter ablation or medical therapy. Patients receiving intervention had better LVEF, peak oxygen consumption and QoL at 6-month follow-up. However, the study was unblinded and conducted at a single centre.[56] In the CAMERA-MRI study,[57] 68 patients with persistent AF and idiopathic cardiomyopathy (LVEF ≤45%) were randomly allocated to catheter ablation or rate control. At 6-month follow-up, cardiac magnetic resonance demonstrated an improved absolute LVEF of 18% ± 13% in the ablation group compared with 4.4% ± 13% in the rate-control group ($P < .0001$) and normalized (LVEF ≥50%) in 58% vs 9% ($P = .0002$).[57] A meta-analysis of six RCTs comparing catheter ablation with medical therapy to medical therapy alone in patients with AF and HFrEF demonstrated a reduction in HFH, CV mortality, and all-cause mortality with ablation as well as improvement in LVEF and QoL.[58]

There is a general lack of RCTs investigating the role of catheter ablation of AF in patients with HFrEF due to arrhythmia-induced cardiomyopathy. The same is true for patients with HFpEF.

A small RCT demonstrated that catheter ablation improved haemodynamic parameters, exercise capacity, and QoL compared with medical therapy in patients with HFpEF.[59] A recent post-hoc analysis of the CABANA trial showed that a high probability of HFpEF may be associated with a greater beneficial effect of catheter ablation as compared to medical therapy on CV hospitalization, but not on all-cause death.[60] However, a meta-analysis of 12 RCTs including patients with AF and HF regardless of LVEF showed that catheter ablation for AF was associated with reduced risk of HF events as compared with conservative care, but had limited or no benefit in HFpEF.[61]

Numerous RCTs on catheter ablation for AF in HFrEF (NCT06560047, NCT06505798) and HFpEF (NCT04160000, NCT06125925, NCT04282850) are ongoing.

6.2. Chronic coronary syndromes

6.2.1. Myocardial revascularization

The STICH trial compared coronary artery bypass grafting (CABG) to medical therapy in patients with CAD amenable to treatment with CABG and an LVEF ≤35%. There was no difference in the primary endpoint of death from any cause at a median of 56-month follow-up.[62] However, the extended follow-up to 10 years showed a significant reduction in death from any cause in the CABG vs medical therapy arm (58.9% vs 66.1%; HR 0.84; 95% CI 0.73 to 0.97; $P = .02$).[63] Secondary analyses of the STICH trial suggested that neither myocardial viability, angina, nor inducible ischaemia were related with outcomes after revascularization.[64–66] Notably, STICH included patients with a mean age of 60 and was conducted before the occurrence of some of the important advances in medical therapy for HFrEF.

The REVIVED-BCIS2 study compared percutaneous coronary intervention (PCI) and optimal medical therapy to optimal medical therapy alone in patients with LVEF ≤35% and extensive CAD amenable to PCI.[67] Median age was 70 years. At a median follow-up of 41 months, there was no difference between the two arms of treatment for the primary composite endpoint of death from any cause or HFH (HR 0.99; 95% CI 0.78 to 1.27; $P = .96$). Secondary analyses of the REVIVED-BCIS2 trial suggested that neither myocardial viability nor completeness of revascularization were related with outcomes after PCI vs medical therapy.[68,69]

Three recent large registries including patients with reduced LVEF undergoing CABG or PCI because of obstructive CAD, reported a possible survival advantage of CABG over PCI up to 5-year follow-up.[70–72] However, these studies were retrospective and non-randomized.

A post-hoc analysis of the ISCHEMIA trial showed more favourable outcomes in the small subgroup of patients with HF and LVEF between 35% and 45% receiving an initial invasive management for chronic coronary syndrome (CCS) (coronary angiography followed by revascularization with PCI or CABG) as compared with those treated conservatively.[73]

In a meta-analysis including five RCTs and 2842 patients with HF and CAD, revascularization with CABG or PCI was found to be beneficial as compared with medical therapy in reducing all-cause death, but with an HR of 0.88 and an upper 95% CI close to 1.[74]

There are no specific RCTs or meta-analyses evaluating the role of myocardial revascularization for outcome improvement in patients with HFpEF.

Regarding patient-reported outcomes, in the STICH trial, CABG led to a significant improvement in QoL as compared with medical therapy over 36 months; in the REVIVED-BCS2 trial, PCI led to significant improvement in QoL as compared with medical therapy at 6 months, but the benefit was not sustained to 1- or 2-year follow-up. Notably, both trials were unblinded.

Improvement of angina after revascularization in patients with CCS is supported by several studies, although the proportion of patients with HF included in these studies is generally limited.[75,76]

6.3. Valvular heart disease

6.3.1. Mitral regurgitation

The three available RCTs comparing mitral transcatheter edge-to-edge repair (M-TEER) + FMT vs FMT alone in patients with HFrEF and significant mitral regurgitation (MR) are: MITRA-FR,[77] COAPT,[78] and RESHAPE-HF2.[79]

MITRA-FR randomized 304 patients to M-TEER and FMT vs FMT alone. The primary endpoint was a composite of all-cause death or unplanned hospitalization for HF at 12 months. No differences between the two arms of treatment were observed [odds ratio (OR) 1.16; 95% CI 0.73 to 1.84; $P = .53$]. Importantly, this trial did not exclude patients with advanced HF, right side involvement, or haemodynamic instability. Also, there was no eligibility committee to evaluate FMT optimization before randomization.[77]

COAPT enrolled 614 patients to M-TEER and FMT vs FMT alone. The primary endpoint was all hospitalizations for HF within 24 months of follow-up. Transcatheter edge-to-edge repair on top of FMT was found to be beneficial for both the primary endpoint (HR 0.53; 95% CI 0.40 to 0.70; $P < .001$) and all-cause death (HR 0.62; 95% CI 0.46 to 0.82; $P < .001$).[78]

RESHAPE-HF2 enrolled 505 patients to M-TEER and FMT vs FMT alone. The three co-primary endpoints were met: the composite of CV death or HFH [rate ratio (RR) 0.64; 95% CI 0.48 to 0.85; $P = .002$], HFH (RR 0.59; 95% CI 0.42 to 0.82; $P = .002$), and Kansas City Cardiomyopathy Questionnaire overall summary score (mean difference, 10.9 points; 95% CI 6.8 to 15.0; $P < .001$). However, there was no difference between the two groups for all-cause death (RR 0.90; 95% CI 0.71 to 1.13; $P = 0.37$). Patients included in the RESHAPE-HF2 had lower effective regurgitant orifice area as compared with the other two trials reflecting the inclusion of patients with moderate to severe MR. Notably, both COAPT and RESHAPE-HF2 excluded patients with advanced HF, right ventricular dysfunction, severe tricuspid regurgitation, and haemodynamic instability.[79]

A meta-analysis of the three M-TEER RCTs[80] showed a relatively large degree of heterogeneity among the trials; however, better outcomes for the M-TEER + FMT group compared with FMT alone in terms of unplanned HFH within 24 months (HR 0.63; 95% CI 0.50 to 0.80; $P = .0002$), first event of hospitalization for HF or all-cause mortality within 24 months (HR 0.72; 95% CI 0.51 to 0.999, $P = .0493$), and change in 6-min walk distance (6MWD) from baseline to 12 months (mean change in metres, 28.42; 95% CI −2.44 to 59.28; $P = .0711$) were observed.[80]

MATTERHORN is a non-inferiority trial, showing that among operable patients with HF, LVEF >20% and symptomatic secondary MR despite FMT, M-TEER was safer and non-inferior to mitral valve surgery with respect to a composite of death, rehospitalization for HF, stroke, reintervention, or implantation of an LVAD at 1 year.[81]

7. Non-cardiovascular comorbidities

7.1. Obesity

The STEP-HFpEF trial enrolled 529 patients with LVEF ≥45%, HF symptoms, a BMI ≥30 kg/m^2, and with no history of diabetes. Participants were randomized to once weekly semaglutide (2.4 mg) or placebo and followed for 52 weeks. Semaglutide led to greater weight loss, larger reductions in symptoms and physical limitations, greater improvements in exercise function than placebo.[82] Similar results were found with semaglutide in patients with HFpEF, obesity, and diabetes in the STEP-HFpEF DM trial.[83]

A prespecified analysis of the SELECT trial, including adults with a BMI of ≥27 kg/m^2 and established CV disease randomized to semaglutide or placebo, examined the effect of semaglutide compared with placebo in patients with and without a history of HF at enrolment. Semaglutide improved all outcome measures, including major adverse cardiovascular events (a composite of non-fatal myocardial infarction, non-fatal stroke, and CV death), the composite HF outcome (CV death or hospitalization or urgent hospital visit for HF), CV death, and all-cause death, in patients with HF and regardless of the HF subtype.[84]

In a post-hoc pooled, participant-level analysis of SELECT, STEP-HFpEF, STEP-HFpEF DM, and FLOW trials, semaglutide reduced the risk of the composite endpoint of CV death or worsening HF events, without an effect on CV mortality alone, in 3743 participants with HFpEF.[85]

The SUMMIT trial[86] found that tirzepatide, a dual agonist of glucose-dependent insulinotropic peptide and glucagon-like peptide-1 receptors, reduced the combined primary endpoint of CV death or a worsening HF event by 38% in 731 patients with HFpEF and a BMI ≥30 kg/m^2 during a median follow-up of 104 weeks. The results were mainly driven by a significant effect on worsening HF events. Tirzepatide was also associated with improved health status, significantly reduced body weight, and improved functional capacity.[86]

7.2. Iron deficiency and anaemia

The AFFIRM-AHF and IRONMAN trials, testing i.v. ferric carboxymaltose and i.v. ferric derisomaltose vs placebo, respectively, in patients with HFrEF and iron deficiency (ID) failed to meet the primary composite endpoint of CV death or total HFH, despite a point estimate suggesting benefit.[87,88] In both trials, prespecified coronavirus disease 2019 (COVID-19) analyses showed a significant risk reduction of the primary endpoint with i.v. iron supplementation compared with placebo.[87]

Furthermore, a significant reduction in total HFHs was observed in AFFIRM-AHF.[87]

In the HEART-FID trial, i.v. ferric carboxymaltose did not reduce the hierarchical composite endpoint of death from any cause or HFH within 12 months, and change in the 6MWD at 6 months in ambulatory patients with HF, an LVEF ≤40%, and ID.[89] Of note, the patient population in HEART-FID was a lower-risk population than the one with acute HF enrolled in the AFFIRM-AHF trial (all patients hospitalized) and the IRONMAN trial (14% of patients enrolled while hospitalized), the mean transferrin saturation (TSAT) was higher in the HEART-FID trial than in previous studies, and dose of ferric carboxymaltose and the timing of its administration also differed from those in the AFFIRM-AHF trial.[89]

In a meta-analysis of 6624 HF patients (predominantly with LVEF <45%), using aggregate-level data, treatment with i.v. iron resulted in a lower risk for CV death, combined CV death and HF admission, first HF admission, and total HF admissions.[90] However, this meta-analysis had a high level of heterogeneity, including large RCTs but also small and old studies with low number of patients and using formulations not studied in HF, such as iron sucrose. Meta-regression suggested that some of the heterogeneity was related to the baseline TSAT of the enrolled population, with trials enrolling patients with lower TSAT exhibiting a large effect size on HF-related events. Similar results, with i.v. iron associated with reduction in the risk of composite of HFH were found in other meta-analyses, that did not include HEART-FID and FAIR-HF2.[91–94]

The most recent FAIR-HF2 trial enrolled 1105 patients with HF, LVEF ≤45%, and ID (serum ferritin level <100 ng/mL; or if TSAT was <20%, a serum ferritin level between 100 ng/mL and 299 ng/mL). Patients were randomized to administration of ferric carboxymaltose (n = 558) initially given at an i.v. dose of up to 2000 mg that was followed by 500 mg every 4 months (unless stopping criteria were met) vs a saline placebo (n = 547). CV death or first HFH (first primary outcome) occurred in 141 in the ferric carboxymaltose group vs 166 in the placebo group (HR 0.79, 95% CI 0.63 to 0.99; P = .04). The rate ratio for the second primary outcome (total HFHs) was 0.80 (95% CI 0.60 to 1.06; P = .12). The third primary outcome (CV death or first HFH in patients with a transferrin saturation <20%) occurred in 103 patients in the ferric carboxymaltose group vs 128 patients in the placebo group (HR 0.79, 95% CI 0.61 to 1.02], P = .07).[95]

A meta-analysis of six trials (FAIR-HF, CONFIRM-HF, AFFIRM-AHF, IRONMAN, HEART-FID and FAIR-HF2), including 7175 patients, showed that patients assigned to i.v. iron, compared with those assigned to placebo, had lower rates for the composite endpoint of recurrent HFH and CV mortality at 12 months [risk ratio (RR) 0.72, 95% CI 0.55 to 0.89] and for the complete length of follow-up (RR 0.81, 95% CI 0.63 to 0.97). The RRs for recurrent HF hospitalizations were 0.69 (95% CI 0.48 to 0.88) and 0.78 (95% CI 0.55 to 0.98) at 12 months and the complete length of follow-up, respectively. CV mortality alone was not significantly reduced (HR 0.80, 95% CI 0.61 to 1.03, and HR 0.87, 95% CI 0.73 to 1.04, respectively).[96]

7.3. Cancer

Table S21 Definition of cancer therapy-related cardiac dysfunction

Category	Severity	Description
Symptomatic CTRCD (HF)[a]	Advanced HF	HF requiring inotropic support, mechanical circulatory support, or consideration of heart transplantation
	Severe HF	HF hospitalization
	Moderate HF	Need for outpatient intensification of diuretic and HF therapy
	Mild HF	Mild HF symptoms, no intensification of therapy required
Asymptomatic CTRCD	Severe	New LVEF reduction to <40%
	Moderate	New LVEF reduction by ≥10% points to an LVEF of 40%–49% OR New LVEF reduction by <10% points to an LVEF of 40%–49% AND either new relative decline in GLS by >15% from baseline OR new rise in cardiac biomarkers[b]
	Mild	• LVEF ≥50% AND new relative decline in GLS by >15% from baseline • AND/OR new rise in cardiac biomarkers[b]

BNP, B-type natriuretic peptide; cTn I/T, cardiac troponin I/T; CTRCD, cancer therapy-related cardiac dysfunction; GLS, global longitudinal strain; HF, heart failure; LVEF, left ventricular ejection fraction; NT-proBNP, N-terminal pro-B-type natriuretic peptide.
[a]Stages C and D HF.
[b]cTn I/cTn T >99th percentile, BNP ≥35 pg/mL, NT-proBNP ≥125 pg/mL, or new significant rise from baseline beyond the biological and analytical variation of the assay used.

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Table S22 Cancer drugs causing heart failure

Cancer therapy	Indication
Anthracycline chemotherapy (doxorubicin, epirubicin, daunorubicin, idarubicin)	Breast cancer, lymphoma, acute leukaemia, sarcoma
HER2-targeted therapies (trastuzumab, pertuzumab, trastuzumab emtansine T-DM1, lapatinib, neratinib, tucatinib)	HER2+ breast cancer HER2+ gastric cancer
VEGF inhibitors TKIs (sunitinib, pazopanib, sorafenib, axitinib, tivozanib, cabozantinib, regorafenib, lenvatinib, vandetinib) and antibodies (bevacizumab, ramucirumab)	VEGF TKIs: renal cancer, hepatocellular cancer, thyroid cancer, colon cancer, sarcoma, GIST Antibodies: breast cancer, ovarian cancer, gastric cancer, gastro-oesophageal cancer, colon cancer
Multi-targeted kinase inhibitors: second and third generation BCR-ABL TKIs (ponatinib, nilotinib, dasatinib, bosutinib)	Chronic myeloid leukaemia
Proteasome inhibitors (carfilzomib, bortezomib, ixazomib) **Immunomodulatory drugs** (lenalidomide, pomalidomide)	Multiple myeloma
Combination RAF and MEK inhibitors (dabrafenib + trametinib, vemurafenib + cobimetinib, encorafenib + binimetinib)	RAF mutant melanoma
Androgen deprivation therapies GnRH agonists (goserelin, leuprorelin) Antiandrogrens (abiraterone)	Prostate cancer, breast cancer
Immune checkpoint inhibitors: anti-programmed cell death 1 inhibitors (nivolumab, pembrolizumab) anti-cytotoxic T-lymphocyte-associated protein 4 inhibitor (ipilimumab) anti-programmed death-ligand 1 inhibitors (avelumab, atezolizumab, durvalumab)	Melanoma (metastatic and adjuvant) Metastatic renal cancer, non-small cell lung cancer, small cell lung cancer, refractory Hodgkin's lymphoma, metastatic triple negative breast cancer, metastatic urothelial cancer, liver cancer, MMR-deficient cancer

GIST, gastro-intestinal stromal tumour; GnRH, gonadotropin-releasing hormone; HER2, human epidermal growth factor receptor 2; MEK, mitogen-activated protein kinase; MMR, mismatch repair; TKI, tyrosine kinase inhibitor; VEGF, vascular endothelial growth factor.
Table from McDonagh *et al.*[97]

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7.4. Anxiety, depression, and cognitive impairment

Several validated questionnaires are available for screening clinically significant depressive symptoms.[98]

In clinical practice, the Patient Health Questionnaire (PHQ)-2 is a widely used, simple, and effective screening tool that assesses depressed mood and anhedonia using a six-point scale.[99] A PHQ-2 score ≥3 is generally considered a positive screen, offering reasonable diagnostic accuracy and prognostic value. Such findings warrant further evaluation with a more comprehensive tool, such as the PHQ-9.[100]

The PHQ-9 expands on the PHQ-2 by assessing multiple domains of depression on a 27-point scale.[101,102]

It can be used to grade the severity of depressive symptoms, aiding in the identification of high-risk individuals who may benefit from timely intervention. In research contexts, a variety of screening tools have been used (Supplementary data online, *Table S23*).

It is important to note that the presence of depressive symptoms—identified through questionnaires—does not necessarily equate to a clinical diagnosis of depression. Many screening instruments include symptoms not typically emphasized in clinical interviews, which may lead to overestimation of depression prevalence due to potential diagnostic bias.

Table S23 Common tools for the screening of depression

Name	No. of items	Time (min)	Scoring	Advantages	Disadvantages
Patient Health Questionnaire-2 (PHQ-2)	2	<1	0 to 3-point scale	Short and simple screening tool	• Self-reporting • Low sensitivity in patients with CAD
Patient Health Questionnaire-9 (PHQ-9)	9	2–5	0 to 3-point scale	• Based on DSM criteria • Could be used for diagnosis and evaluation of symptom severity	Self-reporting
Hospital Anxiety and Depression Scale (HADS)	14 (7 depression, 7 anxiety)	2–5	0 to 3-point Likert scale (0–21)	• Simple and short • Reliable screening tool • Focuses on anhedonia	• Lack of evaluation of physical symptoms, cognitive symptoms, and suicidal ideation • Not ideal for cardiac units
Beck Depression Inventory-II (BDI-II)	21	10–15	0 to 3-point scale	• Simple, clear and short • Covers physical and cognitive symptoms • Could be useful for evaluation of symptom severity • Good test–retest reliability	• Self-reporting • Not suitable as diagnostic tool
Hamilton Rating Scale for Depression (HAM-D)	17 (updated to 21)	15	Either 0 to 2 or 0 to 4-point scale	• Short • Based on DSM criteria • Assesses symptom severity and treatment response	• Test–retest reliability is controversial • Requires trained interviewers • Lack of symptom domains related to depression
Geriatric Depression Scale (GDS)	15 or 30	5–7	0 to 1-point scale (Yes/No)	• Simple • Good for primary care use	• May underestimate depression by omitting somatic symptoms
Cardiac Depression Scale (CDS)	26	5	0 to 3-point scale	• Easy • Valid and reliable cardiac measure for depression • Screening instrument for early identification	• Self-reporting

CAD, coronary artery disease; DSM, Diagnostic and Statistical Manual of Mental Disorders.

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8. Multidisciplinary team management, long-term follow-up, and patient care

8.1. Multidisciplinary management of heart failure

Cardiologist HF care results in greater prescribing of HF therapies, improved outpatient follow-up, and reduced mortality.[103,104] The role of HF nurses is well established and has been shown to reduce HF and all-cause hospitalizations, reduce length of stay, improve health-related QoL, and reduce mortality.[105,106] The inclusion of pharmacists and cardiac physiologists in HF management programmes (HF-MPs) has also shown beneficial effects on hospitalization and HF optimization, respectively.[107,108] People with HF often manage multiple comorbidities and social determinants of health that impact their outcomes.[109–111] Ensuring that HF-MPs incorporate health and social care professionals relevant to the patient's needs is essential. Insights from two surveys have shown that patients and healthcare professionals have the same perception of treatment goals, but patients need individualized care and explanation of the disease.[112]

Home visits and multidisciplinary HF clinics reduced all-cause hospitalization and, with structured telephone support, reduced HFH and mortality. A network meta-analysis evaluated the effectiveness of different models of transitional HF services following an HF diagnosis.[113] It demonstrated that programmes inclusive of nurse-home visits and disease management clinics reduced all-cause mortality and the addition of nurse-case management reduced all-cause readmissions. In the hospital setting, including HF patients admitted to non-cardiology wards, review by HF specialists was associated with a higher rate of implementation of medical therapy, fewer discharges on diuretic therapy, and lower in-hospital and long-term mortality.[103]

The SENECOR study assigned patients over 75 years with a recent HFH to usual care with a cardiologist or follow-up with a cardiologist and geriatrician.[114] Follow-up of patients by a cardiologist with assessment and appropriate interventions from a geriatrician resulted in a reduction in all-cause hospitalization. The need for integrated cardio-renal-metabolic models of care has been described; however, there remains a gap in the evidence describing these models of care.[115,116]

Primary care professionals (PCP) are well-placed to manage comorbidities that contribute to the development of HF, as well as recognizing the signs and symptoms of HF facilitating early referral to an HF-MP.[117] The CHAMP-HF study described strategies to improve collaboration with PCP, including local education initiatives and pathway development to improve awareness of HF guidelines, improvements in HF prescribing, and referrals to HF-MPs.[118] A meta-analysis of the involvement of PCP in HF care demonstrated an improvement in mortality following an HFH.[119]

8.2. Patient education, self-care, and lifestyle advice

A meta-synthesis identified that managing multiple conditions, severity of illness, insufficient knowledge and subsequent application of that knowledge, length of educational programmes, social isolation, poor healthcare professional–patient communication, psychological distress, and cultural beliefs were barriers to HF self-care.[120] Limitations of studies on self-care include diversity in the interventions, delivery by specialist professionals and not patient-driven, and lack of patient participation in their design.[121] Strategies to improve self-care include co-design of self-care programmes, inclusion of patients in the design of HF education, effective communication, and understanding of barriers specific to individual patients. Signposting to resources such as www.heartfailurematters.org and other third-sector organizations can provide education and support to patients and their caregivers.

Table S24 Key components of heart failure and self-care education

Overarching principle: **The cornerstone of HF education is shared decision-making, facilitating a partnership between patients, their caregivers, and healthcare professionals to develop individualized advice, guidance, and treatment decisions.**		
Education topic	**Goals for the patient and caregiver**	**Healthcare professional role** **Advice, assessment tools, improving self-care adherence**
Understanding HF	To understand HF, its cause, signs and symptoms, and treatment options specific to you.	Provide information tailored to the patient's individual presentation, cause of HF, assessment and management strategies. Consider health literacy when tailoring information.
Comorbidities and their interaction with HF	To identify comorbidities and understand their relationship to HF either as a cause or contributor to HF. To understand the benefits of optimization of comorbidities.	Identify comorbidities that impact HF, response to treatments and follow-up. Work with the MDT to provide holistic HF care in the presence of comorbidities.

Continued

Follow-up	Recognize the aims of follow-up for HF and other health conditions. Be able to identify a change in signs or symptoms that may trigger earlier HF review.	Explain the rationale for long-term follow-up including clinical review, blood tests, ECG and imaging. Provide advice on when to trigger HF review.
The HF trajectory	To understand the HF trajectory and the different phases of HF. To understand management options at different stages to be able to make joint treatment decisions.	Sensitively communicate information on the stages of HF and expectations. Facilitate advanced care planning, palliative care and end-of-life discussion and decisions when appropriate. Identify emotional and practical concerns, utilizing the MDT to address these.
Medical treatment		
Medication	To understand how HF medications work, why they are prescribed, the duration of therapies and common side effects. Be able to describe the importance of adhering to a medication regime and how to communicate with healthcare professionals if you experience side effects or face difficulties in taking medications. Understand the effects of diuretics and how to adjust timing of administration to facilitate lifestyle. Describe what to do if you miss a dose of medication or are fasting, e.g. before a procedure or due to vomiting or diarrhoea. To have enough information to make decisions following advice from healthcare professionals.	Describe medication indication, evidence base, mechanism of action, duration, timing, side effects and reporting. Explain how to manage missed doses of medications and management during fasting or illness. Use different communication techniques to support understanding including oral, written, and digital resources. Identify people who require support aids, e.g. dosette box or delivery of medications.
CIED, e.g. pacemaker, defibrillator, cardiac resynchronization therapy	To understand the reason for CIED implantation, the evidence supporting the recommendation, possible risks and benefits to help you to make an informed decision. Describe follow-up of CIED and how to manage new complications. Understand implications for driving in your country. Understand the management of the CIED at end of life and opportunities for decision-making at that time.	Describe the indication, evidence-base, risks and benefits of CIED. Explain procedure, follow-up and management of potential complications. Use different communication techniques to support understanding including oral, written, and digital resources. Describe lifestyle implications and when to alert national driving agency. Early in CIED decision-making, discuss options for later in HF trajectory to alleviate concerns about CIED (especially defibrillators) at end of life.
Self-care aspects		
Symptom monitoring and self-management	Understand how to maintain self-care through behaviours in the sections below. Describe HF signs, symptoms, and changes to monitor. Identify strategies to support regular monitoring and self-assessment of HF symptoms. Know how to respond to changes in symptoms and when to escalate to a healthcare professional.	Provide information on HF signs and symptoms, monitoring and responding to changes. Support self-management of a change in symptoms, e.g. altering diuretic dose in response to increased breathlessness and/or sudden unexpected weight gain, ensuring patients and caregivers know when to seek review from healthcare professionals.
Physical activity and exercise	Understand the benefits of physical activity and exercise in HF. To recognize when physical activity or exercise needs to be adapted depending on HF symptoms or other health problems. Speak to a caregiver or healthcare professional if there is anxiety about re-starting or increasing exercise.	Provide general information about the benefits of physical activity and exercise in HF. Discuss individualized exercise recommendations and information on how to adapt depending on symptoms. Discuss referral to a cardiac rehabilitation or exercise programme.

Continued

	Understand limitations that may be placed on exercise depending on the cause of HF.	
Sleeping and breathing	Understand why a good sleeping pattern is good for HF and general health. Identify problems with sleep and strategies to manage them. Recognize when symptoms of HF are a cause for sleep disruption and when to alert a healthcare professional.	As part of routine assessment, ask about sleeping pattern and sleep disturbance including feedback from the partner. Explain the benefits of sleep and strategies to improve sleep. Address any concerns relating to the impact of medications on sleep. If there are concerns about sleep-disordered breathing, discuss referral for sleep studies with patient.
Fluids	Understand the principles of congestion in HF and how to manage. If it is advised, adhere to a fluid restriction. Understand the importance of avoiding dehydration. When fasting for a procedure or for religious reasons know how to manage HF medications and diuretics.	Explain the principles of congestion in HF. Routine fluid restriction is not advised for all HF patients; however, it may be appropriate during times of decompensation and any advice should be tailored to the individual. Describe adaptations to fluid intake during periods of hot weather, diarrhoea, vomiting, and increases in weight related to congestion.
Healthy diet	Understand the importance of a stable, healthy diet. Seek advice if unsure about how to maintain a healthy diet or if you notice significant weight loss or gain.	Assess food and supplement intake and ask about unintended weight loss. Facilitate access to advice on healthy eating. Routine restriction of salt intake is not advised for all HF patients. Where appropriate and with consent from the patient, refer to dietician. For patients with obesity facilitate access to advice, support, and other therapies as appropriate.
Alcohol	Avoid heavy alcohol consumption and follow guidance on government-recommended drinking guidelines. Where the cause of HF is related to excessive alcohol consumption, understand why abstinence is advised.	Provide individualized advice regarding alcohol intake, especially in cases of alcohol-related cardiomyopathy. Reinforce government guidelines for alcohol consumption.
Immunization	Understand the reason for the recommendation of immunizations for influenza, pneumococcal, coronavirus, and travel vaccines. Express any concerns to a healthcare professional to support informed decision-making.	Provide information on the rationale for immunizations and signpost to accurate information on the benefits and risks. Respect decision-making relating to immunization.
Smoking and recreational drugs	Describe the harms associated with smoking generally and specific to HF. Seek support to stop smoking including through healthcare professionals, caregivers and friends and peer support. As a caregiver consider support you may need to also stop smoking. Understand the risks of recreational drugs and seek support to stop their use.	Provide information on the health implications of smoking. Following discussion with the patient, refer to smoking cessation services. Encourage caregivers to support smoking or recreational drug cessation including accessing support for them too.
Travel, leisure, driving	Know how to prepare for travel and considerations relating to travelling with HF and how this may impact mode of transportation, destination, and duration of travel. Be aware of practical considerations including taking extra medication in case of travel delays, use hand	Provide individualized advice on travelling taking into account current symptoms and change in symptom profile. Signpost to information to support travelling with HF including awareness of managing hot and cold climates, long travel times, and altitude.

Continued

	luggage for medications and any extra support required while travelling. Identify vaccines relevant to travel destination. Understand the impact of altitude and dehydration on HF symptoms, how to avoid and manage. Identify measures of sun protection including medications associated with photosensitivity. Know what extra considerations are required if travelling with a CIED. Be aware of and follow driving regulations in country or residence and country of travel.	Provide specific advice on monitoring fluid balance and managing diuretics. Highlight national driving regulations and the importance of being alert to change in symptoms that may impact ability to drive safety.
Sexual and reproductive health	Be aware of the impact of HF, associated anxiety or depression and the impact of some medications (e.g. beta-blockers) on sexual desire and function. To feel comfortable and confident resuming sexual activity with HF and strategies to support. Understand symptoms of the menopause, its impact on HF symptoms, and have information to make informed choices relating to management of menopause. For women of reproductive age, understand the implications of pregnancy with HF and have information to make informed choices relating to contraceptive choices. Be aware of the risk of inheritance of genetic causes of HF and have information to support choices related to preimplantation genetic testing and reproductive therapies.	Provide information to support sexual activity including strategies to manage anxiety, erectile dysfunction, symptoms on exercise among others. Provide information and signpost women of menopausal age with HF to strategies to manage menopausal symptoms. Facilitate pre-conception care and counselling for men and women of reproductive age with HF including advice regarding pregnancy with HF, effective contraception and information on assisted reproductive therapies and preimplantation genetic testing.
Living with HF		
Psychological issues	Understand the interaction of HF and mental and emotional well-being on patient and caregivers. Identify the impact mental and emotional well-being has on ability to manage HF and provide self-care. To recognize when to seek advice and support regarding a change in mental health. Accept and understand the potential barriers in continuation of labour market connection.	Describe to patients and caregivers the impact of HF on mental and emotional well-being and the impact on self-care and adherence. Provide information to help recognize mental and emotional distress and how to seek support. Screen for psychological distress using a validated tool and refer appropriately for further assessment and/or management. Provide information on physical and mental barriers for continuing on the labour market.
Family and informal caregivers	Understand HF, recognizing signs and symptoms, its management, and when to seek advice from a healthcare professional. Recognize the important role of caregiver in HF care. Identify your needs to be able to support HF care. Know when and how to seek additional advice or support from health and social care professionals or peer support groups.	Recognize the important role of the caregiver in HF care and decision-making. With the patient's consent, provide information and advice to support HF management and self-care. Provide tools to assist the caregiver in accessing additional information and support.
Tools to support information sharing		
HeartFailureMatters ESC patient guidelines Third-sector organizations Visual aids Animations Information leaflets Peer groups/support Teachback—AskTellAsk		

CIED, cardiac implantable electronic device; ECG, electrocardiogram; ESC, European Society of Cardiology; HF, heart failure; MDT, multidisciplinary team.

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8.3. Implementation and adherence to therapy

Table S25 Barriers and facilitators to improved implementation and adherence

	Barrier	Facilitator
Healthcare professional implementation		
Evolving HF landscape	Unfamiliarity with contemporary evidence for HF management and treatments.	Access HF-focused education. Awareness of HF guidelines and management protocols. Collaboration with HF specialists.
Healthcare system and resources	Lack of access to an HF-MDT. Delays in assessment and optimization of therapies due to limited or ineffective resources. Financial limitations, e.g. limited insurance coverage, low-income healthcare system. Lack of robust audit processes to identify implementation of guidelines and report outcomes.	Development of local guidance and integrated care pathways that includes doctors, nurses, pharmacists, healthcare scientists, and allied professionals. Advocacy for improved affordability and reimbursement. Development of regular feedback and audit mechanisms to support adherence to HF guidelines.
Comorbidities, e.g. frailty, cognitive impairment	Uncertainty when managing HF alongside common chronic conditions, e.g. diabetes, CKD, chronic obstructive pulmonary disease. Difficulty prioritizing therapies across multiple conditions.	Multidisciplinary working to develop integrated care plans for multimorbidity management. Access to multidisciplinary teams to address complex cases.
Polypharmacy	Concern regarding potential drug interaction. Uncertainty of current medications.	Medicine reconciliation and pharmacist review.
Clinical inertia	Perception of HF 'stability'. Uncertainty of evidence of new therapies and adoption of a 'wait-and-see' approach. Dependence on established clinical practices.	Recognition of ongoing risk of adverse clinical events for HF patients. Review of contemporary HF guidelines. Development of supportive tools, e.g. titration protocols and reminder systems.
Unconscious bias	Unconscious biases resulting in inequality in access to HF treatment for specific groups of patients, e.g. gender, race, ethnicity, low-income.	Reflect on own practice and identify potential biases. Seek feedback from patients and colleagues. Access training to identify and reduce bias. Use HF guideline to support standardized approach to management.
Patient/caregiver adherence		
Knowledge and understanding	Limited understanding of HF, signs and symptoms, treatments, and self-care. Misconceptions about HF and its treatments. Mixed messaging from healthcare professionals.	Tailored education and use of patient-centred, plain language materials, e.g. animations, leaflets, and inclusion of caregivers in their native language. Identify patient's ideas, concerns and expectations, and provide information to address concerns. Clear documentation for each consultation and structured approach to care to reduce inconsistencies in information and management.
Health literacy	Difficulties communicating with healthcare professionals. Readability and accessibility of educational resources. Challenges accessing healthcare services.	Assess health literacy. Use plain language tools and visual aids aimed at a reading age around 9–11 years. Use teach-back techniques to assess understanding and areas that need further explanation. Provide opportunities for patients and caregivers to ask questions. Support access to healthcare by adapting clinic venue and times, optimizing review with other healthcare professionals simultaneously.

Continued

Social determinants of health	Financial barriers: • out-of-pocket expenses to attend appointments • insurance costs, low reimbursement • access to affordable healthcare services • job security • food insecurity • housing • social isolation	Signpost to financial assistance programmes and financial counselling. Identify areas in local HF pathways that could be streamlined or delivered differently to improve access. Recognize the impact of education on HF self-care and management and support access to information and advice appropriate to patient. Recognize the impact of HF on job security and support patients with information for employers and an individualized follow-up plan to minimize time off from work. Ensure diet and nutrition are core components of HF management and using a validated screening tool, identify food insecurity, and signpost to relevant services. Support patients to access housing suitable for them by providing information to local authorities when required. Identify social isolation and signpost to peer support groups and other supportive service, e.g. home care.
Psychological distress (e.g. anxiety, depression)	Lack of routine assessment of the presence of psychological distress in HF care leads to under-recognition. Lack of understanding on the interaction between HF and psychological distress. Scepticism about therapeutic approaches for psychological distress.	Using a validated tool screen for psychological distress. Utilize the wider multidisciplinary team to facilitate access to management strategies and treatments. Provide advice to patients and caregivers on the impact of psychological distress on HF self-care and strategies to optimize. Explain the benefits of therapies to improve quality of life and HF outcomes.
Cognitive impairment	Difficulty recognizing and diagnosis cognitive issues that affect adherence and understanding. Challenges in creating care plans and follow-up in the presence of cognitive impairment.	Screen for cognitive impairment. Refer to memory or psychiatry services as appropriate. Multidisciplinary collaboration to develop care plans cognizant of the presence of cognitive impairment. Identify additional need for home care or other social support mechanisms. Involve caregivers in needs assessment and decision-making.
Comorbidities	Difficulty understanding interactions between health conditions and assessment of multiple symptoms. Challenges managing multiple healthcare interactions and resulting frustrations.	Identify opportunities for polypharmacy review. Facilitate education and advice on the management of multiple health conditions and recognition of when to seek advice for change in symptoms. Ensure clear management and follow-up plans are in place optimizing the wider MDT.
Digital technology	Lack of access to digital services may limit options to implement HF management.	Support access to digital technology. Ensure alternative strategies to digital follow-up are available.
Polypharmacy	Burden of multiple medications and associated monitoring. Difficulties in managing medication regimes.	Medicine reconciliation and pharmacist review to develop simplified dosing regimes and appropriate de-prescribing. Use of pill organizers or reminder apps

CKD, chronic kidney disease; HF, heart failure; MDT, multidisciplinary team.

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8.4. Exercise training and cardiac rehabilitation

8.4.1. Benefits of exercise training according to heart failure stage and phenotype

The ExTraMATCH II meta-analysis, combining data from 3990 patients and 13 randomized trials, including HF-ACTION, showed consistent benefits in reducing hospitalizations and improving QoL in HFrEF across various subgroups, including age, sex, and NYHA class.[122]

In a diverse population of older patients who were hospitalized for acute decompensated HF, an early, transitional, tailored, progressive rehabilitation intervention resulted in greater improvement in physical function than usual care. This intervention included multiple physical function domains (strength, balance, mobility, and endurance) initiated during or early after hospitalization for HF, and continued after discharge.[123]

8.4.2. Exercise prescription

Moderate-intensity aerobic exercise is the standard; however, recent studies indicate that high-intensity interval training may provide similar improvements in peak oxygen consumption in HFrEF and HFpEF. High-intensity interval training requires close monitoring initially but may allow quicker improvements in functional capacity than moderate continuous training and is a safe option in selected patients. Apart from the traditional mode centre-based cardiac rehabilitation,[124] more innovative delivery models [home-based, digitally supported, and hybrid (mix of home and centre)] may improve participation and appear similarly effective as centre-based cardiac rehabilitation.[125] In fact, 10 of 15 of the new randomized trials identified in a recent Cochrane review were either home-based (five trials) or hybrid (five trials) and reported similar improvements in outcomes as centre-based trials.[126] This finding is supported by a small number of head-to-head trials and recent network meta-analysis that show similar improvements in patient-reported outcomes between centre- and home-based cardiac rehabilitation (with or without digital technology support) in populations with HF.[127,128]

Adherence to the prescribed exercise regimen is crucial.[129] According to the HF-ACTION trial, the largest reductions in HFH were seen in patients with the highest adherence rate.[130] Shared decision-making, personalized exercise prescription based on the patient's preferences (self-selected training) and abilities (age, concomitant diseases, leisure and working habits, logistical restraints) is recommended to increase long-term adherence.[129] In addition, smartphone applications and wearable activity trackers may assist in long-term adherence to physical activity goals and exercise.[131,132]

8.5. Telemonitoring

In addition to the efficacy of remote monitoring modalities, the workload of each modality and the number of false positive alarms vary, which confines any overarching efficacy conclusions. Telemonitoring should be adapted to work in synergy with existing healthcare provision, which makes findings system and setting dependent.[133,134] Strategies that focus on continuous optimization of care rather than trying to anticipate and prevent episodes of worsening HF appear to be more successful, which may be related to the applied parameters that increase later in the period when the patients decompensate.[135]

9. Specific conditions

9.1. Pregnancy and heart failure

9.1.1. Pre-existing stage B or stage C heart failure

9.1.1.1. Pre-conception counselling and risk assessment

Women with structural heart disease, stage B HF, or pre-existing HF have a higher risk of pregnancy-related CV complications, including HF decompensation. In these patients, pre-conception counselling and risk assessment using modified World Health Organization (mWHO) 2.0 should be integrated into routine follow-up to promote informed decision-making regarding contraception, conception, pregnancy, and HF treatments.[136]

9.1.1.2. Pregnancy planning: medical management

For women pursuing pregnancy, medications should be reviewed to prevent harm to the foetus. This must be balanced against the impact of withdrawal of disease-modifying therapies in addition to the haemodynamic burden of pregnancy and should be discussed with the patient. Angiotensin-converting enzyme inhibitors, angiotensin II receptor blockers, ARNIs, MRAs, ivabradine, and sodium–glucose co-transporter 2 inhibitors are contraindicated in pregnancy.[136–139] Withdrawal of these therapies prior to conception should be done under close monitoring, including imaging of ventricular function, natriuretic peptides, and clinical review. In pregnant women with HFrEF, it is recommended that non-selective beta-blockers are switched to beta-1-selective blockers (metoprolol succinate, bisoprolol) with close maternal and foetal monitoring.[140,141] Further guidance is available in the *2025 ESC Guidelines for the management of cardiovascular disease and pregnancy*.[136]

9.1.1.3. Pregnancy planning: device

For individuals with cardiac implantable devices, pre-pregnancy assessments for arrhythmic risk and battery longevity are essential.

9.1.1.4. Pregnancy planning: genetic counselling and fertility treatments

Both men and women with HF should be counselled on the risk of recurrence of inherited cardiac conditions in their offspring and referred for genetic counselling where appropriate.[142,143] Conversations about risk and prognosis for women and men with advanced HF considering parenthood is advised.

Women and men with HF considering fertility treatments should be assessed by a specialized Pregnancy Heart Team for risks associated with fertility treatments and subsequent pregnancy, and explore pre-implantation genetic testing where appropriate.[143,144]

9.1.2. Antenatal, intrapartum, and post-partum care

9.1.2.1. Diagnosis and epidemiology

Physiological changes in pregnancy, including increased blood volume and cardiac output, can lead to a deterioration in cardiac function for women with known HF or unknown pre-disposition to HF, leading to a first HF presentation in pregnancy or post-partum.

The differential diagnoses of HF in pregnancy are broad, and include Takotsubo syndrome or stress cardiomyopathy, peripartum cardiomyopathy (PPCM), exacerbation of a pre-existing cardiomyopathy, and valvular or congenital heart disease.[138,145] HF secondary to a pre-existing cardiomyopathy is more likely to present in the second trimester.[138,146] PPCM is the leading cause of HF in pregnancy and usually presents in the third trimester or in the early post-partum period, but may present much later.[147] The prevalence of PPCM ranges from 1:100 in Nigeria to 1:1000 in South Africa and 1:1500 in Germany.[146] The majority of cases are diagnosed post-partum. Six-month mortality is recorded to be 6% overall, lowest in Europe (4%) and highest in the Middle East (10%).[148] Most deaths are due to HF (42%) or SCD (30%).[148] Cardiac recovery may occur in the first 3–6 months, although it may be delayed to up to 2 years. Recovery rates vary among regions, from 75% to less than 50%.[146,148–150]

9.1.2.2. Monitoring

Frequent monitoring of maternal cardiac function with clinical review, echocardiogram, natriuretic peptides, foetal growth assessment, and routine prevention of pregnancy-associated complications is essential to detect early signs of worsening and prevent adverse outcomes. Adjustments of medical therapies may be required while balancing maternal cardiac function with foetal safety. Educating patients on HF symptoms and emergency response is crucial.

9.1.2.3. Labour and delivery

Plans should be individualized, considering maternal cardiac stability, and the safest mode and setting for delivery. A Pregnancy Heart Team involving obstetricians, cardiologists, anaesthesiologists, and midwives should co-ordinate to ensure appropriate monitoring and intervention options. Planning for treatment of potential complications and escalation of care enhances both maternal and foetal outcomes.

9.1.2.4. Termination of pregnancy

Women with advanced HF who are pregnant and assessed at intermediate/high mWHO 2.0 risk should be referred to a specialist centre for counselling regarding any consideration of termination of pregnancy due to significantly high risks to maternal health. This has to take into account national laws. A multidisciplinary team should facilitate comprehensive counselling to support the woman's decision-making and ensure the safest possible outcome.

9.1.3. Heart failure management during pregnancy

Assessment and management of HF in pregnancy depends on the severity and clinical setting. A thorough evaluation including specialist medical review, echocardiography, electrocardiography, and natriuretic peptides, along with foetal monitoring is required. Additional investigations can be considered in *de novo* presentations or where diagnosis is unclear.

Pregnant women presenting with signs of decompensated HF require urgent hospital admission and assessment by a multidisciplinary Pregnancy Heart Team. Consideration of the most appropriate location for treatment and, eventually, delivery is based on the expertise available and interventions that may be required. Milder cases can be treated with diuretics, beta-blockers, hydralazine, and oral nitrates. Severe cases, such as those presenting with HF and cardiogenic shock, may need transfer to advanced HF centres for interventions with mechanical circulatory support (MCS) or heart transplantation. In unstable HF, MCS or inotropes are required.[151] Inotropes agents are generally considered safe.[137,151] Urgent delivery is sometimes necessary. Bromocriptine, a prolactin inhibitor, may be considered for PPCM to improve heart function, although it increases deep vein thrombosis risk, necessitating concurrent anticoagulation.[136,152–154] An ongoing RCT, the REBIRTH trial, is further investigating the impact of bromocriptine on clinical outcomes in patients with PPCM (NCT05180773).

9.1.3.1. Post-partum

The risk of HF decompensation and mortality continues in the weeks and months following delivery.[150] Due to reports of increased incidence of left ventricular thrombi and systemic thromboembolism in women with PPCM and the hypercoagulable state of pregnancy and the early post-partum period, anticoagulation should be considered for those with LVEF <35% during late pregnancy and post-partum.[149,155] Close monitoring of cardiac function, symptoms, and initiation and optimization of HF medical therapies is essential. Informed decision-making relating to breastfeeding and the concurrent use of HF therapies requires careful counselling to support women to continue with treatments. The contraceptive plan discussed during the antenatal period should be implemented prior to discharge. Follow-up with cardiologists, HF nurses, and PCPs is important to improve outcomes. Referral for additional diagnostics to identify HF aetiology in *de novo* presentations, including cardiac magnetic resonance imaging and genetic counselling should be considered.[142]

9.1.4. Living with heart failure during and after pregnancy

Managing a high-risk pregnancy or a new HF diagnosis often brings emotional strain, impacting mental well-being. Patients and their families may experience anxiety and fear regarding their health and future. Access to the HF-MDT facilitates HF education and advice to support living with HF. Assessment of psychological distress is important to facilitate access to mental health professionals who can help manage psychological distress and promote well-being during this challenging time.[156]

Table S26 Core components of heart failure care in pregnancy: pre-conception to post-partum

Pre-conception risk assessment and stratification	Clinical and heart failure assessment: • Comprehensive clinical review, including medical, family, and obstetric history • Up-to-date diagnostics (ECG, echocardiography, NT-proBNP, others as appropriate) • Functional assessment (NYHA class, exercise capacity) • Medication review (teratogenicity, optimization of therapy, and plan for adjustment during pregnancy) • Routine pre-conception optimization: ensure up-to-date vaccinations, folic acid supplementation, and management of modifiable risk factors • Risk stratification: • Apply mWHO 2.0 class stratification and identify high-risk cardiac features • Evaluate comorbidities and social determinants of health • Assess impact of any necessary medication changes
Shared decision-making and pregnancy planning	• Maternal and foetal risk counselling, with information on heart failure and pregnancy outcomes and education on signs and symptoms to monitor for • Lifestyle modification and pre-conception health optimization • Assisted conception considerations and genetic testing/counselling where indicated • Medication review with plan for adjustment and monitoring • Involvement of partner/family in counselling • Documentation of discussions, agreed pregnancy plan, and advance care preferences where relevant • Access to supporting information and peer or psychological support
Multidisciplinary care	• Collaborative approach across all phases of care • Active input from heart failure specialists within the Pregnancy Heart Team • Involvement of health and social care professionals where relevant • Coordination between tertiary and local centres with clear communication plans • Development of a coordinated, individualized care plan
Antenatal care	• Regular multidisciplinary Pregnancy Heart Team review • Routine antenatal care plus additional heart failure monitoring • Frequency of follow-up based on risk and clinical course • Optimize blood pressure control and manage anaemia and iron deficiency • Medication review and adjustment as required • Foetal growth and well-being monitoring • Advice on activity levels and safe exercise • Consider thromboprophylaxis based on antenatal and post-natal risk assessment
Decompensated heart failure management	• Early recognition of decompensated heart failure symptoms and signs • Prompt assessment, stabilization and relief of congestion • Identify and treat reversible causes • Early involvement of senior clinicians and the Pregnancy Heart Team • Escalate to HDU/ICU or advanced heart failure centre if unstable or requiring advanced monitoring • Careful fluid management and diuretic strategy • Continuous monitoring of maternal and foetal condition • Plan for delivery timing and location if deterioration occurs
Intrapartum care	Develop a dynamic, multidisciplinary delivery plan by 28–32 weeks, addressing: • Planned timing and location of delivery (tertiary centre if high risk) • Foetal concerns and obstetric indications • Mode of delivery • Anaesthetic considerations • Continuous ECG and haemodynamic monitoring as indicated • Escalation plan in event of maternal decompensation • Post-natal care location and monitoring duration
Post-partum care	• Ongoing haemodynamic monitoring as appropriate to clinical risk • Vigilance for de novo or worsening heart failure • Aetiology work-up for *de novo* heart failure presentations • Early post-partum echocardiography and clinical review • Assessment of medication compatibility with breastfeeding

Continued

	• Optimization of heart failure therapy and device management • Family planning and contraception counselling • Psychological support and screening for postnatal depression or anxiety • Early post-partum cardiology review

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ECG, electrocardiogram; HDU, high-dependency unit; ICU, intensive care unit; mWHO, modified World Health Organization; NT-proBNP, N-terminal pro-B-type natriuretic peptide; NYHA, New York Heart Association.

9.2. Cardiomyopathies

9.2.1. Hypertrophic cardiomyopathy

Most people with hypertrophic cardiomyopathy (HCM) are asymptomatic and have a normal lifespan, but some develop symptoms of HF, often many years after the appearance of electrocardiographic or echocardiographic evidence of left ventricular hypertrophy. Assessment of left ventricular outflow tract obstruction (LVOTO) should be part of the routine evaluation of all symptomatic patients. In symptomatic patients with LVOTO, the aim is to improve symptoms by using drugs, surgery, or alcohol septal ablation. For the detailed management of LVOTO, please refer to the *2023 ESC guidelines for the management of cardiomyopathies*.[157] A new trial reinforcing the evidence for the use of cardiac myosin ATPase inhibitors in patients with symptomatic LVOTO and without left ventricular systolic dysfunction has been published. SEQUOIA-HCM is a phase 3, double-blind trial that randomly assigned adults with symptomatic obstructive HCM (NYHA classes II or III, along with decreased exercise capacity) and LVEF ≥60% to receive aficamten (starting dose, 5 mg; maximum dose, 20 mg) or placebo for 24 weeks, with dose adjustment based on echocardiography results.[158] Treatment with aficamten resulted in a significantly greater improvement in peak oxygen uptake than placebo.[158]

Therapy in symptomatic patients without LVOTO focuses on management of arrhythmia, reduction of left ventricular filling pressures, and treatment of angina.

During the trajectory of the disease, about 8% of patients with HCM develop systolic dysfunction with LVEF <50% (7.5% incidence over 15 years in a large retrospective study).[159] The natural history of HCM with left ventricular systolic dysfunction is variable.[159,160] In the large international SHaRe registry, major adverse events (death, cardiac transplantation, or LVAD implantation) occurred in 35% of patients at an estimated median follow-up of 8.4 years.[159] FMT is indicated in patients with HCM developing left ventricular systolic dysfunction. Patients with end-stage systolic or diastolic dysfunction refractory to medical therapy may be candidates for cardiac transplantation.

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